

Galaxy Image Classification and the Galaxy Merger Rate

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We train a convolutional neural network (CNN) to predict Galaxy Zoo 2 (GZ2) morphological vote fractions from Sloan Digital Sky Survey (SDSS) *gri* images. Starting from 61,578 labelled galaxies with 37 vote-fraction labels in $[0, 1]$, we centre-crop and resize the 424×424 cutouts to 96×96 pixels, a $\sim 20\times$ reduction in pixel count, and split 80/20 into 49,262 training and 12,316 validation images. A mean-prediction baseline achieves a validation RMSE of 0.1639. We compare two backbones, a 4-block custom CNN with dropout and ~ 2.8 M parameters and an 18-layer ResNet adapted for small inputs and sigmoid output with ~ 11.2 M parameters, and apply two optimisations: **ReduceLROnPlateau** learning-rate scheduling and random $[0, 360^\circ)$ rotation augmentation exploiting the rotational invariance of galaxy morphology. The augmented, LR-scheduled custom CNN achieves the best validation RMSE of 0.0913, a 44% improvement over the baseline. Per-label scatter plots show that top-level smooth/disk features are predicted accurately, while residuals concentrate in the rare odd-feature leaves (rings, mergers, lens/arcs). We apply the trained model to 79,975 held-out SDSS images and recover a merger fraction $f_{\text{merger}} \approx 4\%$, which converts to a major-merger rate $R_{\text{merger}} \approx 0.04\text{--}0.20 \text{ Gyr}^{-1}$ using the observability-timescale calibrations of J. M. Lotz et al. (2008, 2011) and V. Rodriguez-Gomez et al. (2015). This agrees with the $z \approx 0.1$ Lotz value of $\sim 0.02 \text{ Gyr}^{-1}$ within a factor of ~ 2 , with the morphological visibility timescale $\tau_{\text{vis}} \approx 0.2\text{--}1.1 \text{ Gyr}$ dominating over statistical uncertainty.

1. Introduction

Galaxy morphology is the most direct observational tracer of a galaxy’s dynamics and history (C. J. Conselice 2014). Over large samples, morphology allows us to reconstruct the statistical evolution of the galaxy population without following individual galaxies through cosmic time. In the hierarchical Λ CDM model, the primary mechanism evolving galaxies along its evolutionary sequence are major mergers, where they perturb stellar orbits, drive gas to galactic centres fuelling starbursts and AGN, and convert rotationally-supported discs into pressure-supported spheroids. The major-merger rate as a function of redshift therefore sets the rate at which the early-Universe disc population evolves into the present-day red sequence. Constraining this rate observationally is required for a quantitative theory of galaxy evolution. Locally, this rate has been constrained at the $\sim 0.02\text{--}0.03 \text{ Gyr}^{-1}$ level by observation (J. M. Lotz et al. 2011) and simulation (V. Rodriguez-Gomez et al. 2015), but the two methodologies disagree on the $(1+z)$ evolution slope by roughly a factor of two.

The largest hand-labelled morphological catalogue available is Galaxy Zoo 2 (GZ2; K. W. Willett et al. 2013), comprising $\sim 3 \times 10^5$ SDSS galaxies labelled by citizen-scientist consensus on a 37-leaf decision tree. However, as this labelling pipeline struggle to scale to the LSST, Euclid, and Roman data releases with its reliance on voluntary participation, convolutional neural networks (CNNs) provide a programmatic alternative where given paired images and labels, a CNN learns to predict morphological vote fractions directly

from pixels (A. Krizhevsky et al. 2017).

In this work, we (1) assemble a 61,578-galaxy GZ2 training set with 96×96 pixel SDSS *gri* cutouts and 37 vote-fraction labels per galaxy; (2) train two CNN architectures, an 18-layer ResNet (~ 11.2 M parameters; K. He et al. 2015) and a custom 4-block CNN (~ 2.8 M parameters); (3) compare two optimisations (learning-rate scheduling and 360° rotation augmentation) and one architectural regularisation (tree-reweighted output) against a mean-prediction baseline; (4) apply the best model to a held-out test sample of 79,975 SDSS galaxies; and (5) recover a major-merger rate R_{major} from the predicted Class8.6 merger fraction using the J. M. Lotz et al. (2011) visibility-timescale calibration.

2. Background

2.1. The Hubble Sequence

The Hubble sequence (Figure 1) separates galaxies into two physically distinct branches: pressure-supported ellipticals (E0–E7) and lenticulars (S0), and rotationally-supported spirals (Sa–Sd) and irregulars (Irr). The Sa–Sd sequence within the disk class is ordered by decreasing bulge-to-total ratio (B/T), increasing gas fraction, and increasing star-formation rate per unit mass. Ellipticals are thought to form through major mergers that scramble ordered rotation into a velocity-dispersion-supported spheroid, while disk galaxies assemble more quiescently through cold gas accretion and angular-momentum conservation (C. J. Conselice 2014).

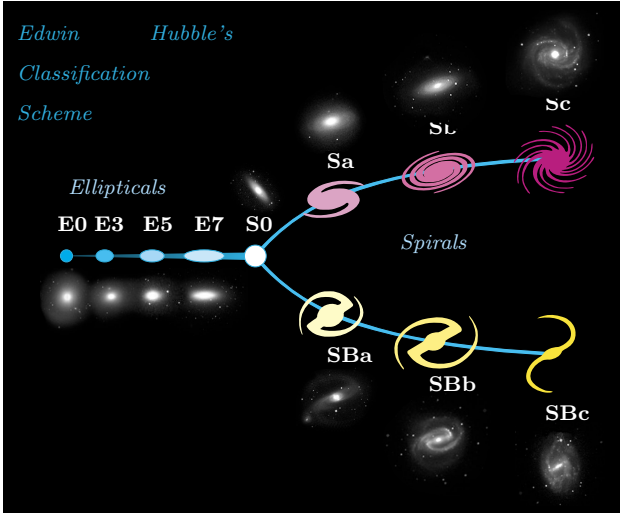


Fig. 1.— The Hubble tuning-fork diagram. Ellipticals (E0–E7) lie along the handle, ordered by apparent flattening. The two prongs separate barred (SBa–SBc) from unbarred (Sa–Sc) spirals along an axis of decreasing bulge prominence and increasingly open arms. Lenticulars (S0) bridge the elliptical and disk families. Image credit: NASA & ESA, <https://esahubble.org/images/heic9902o/>.

2.2. The Cosmic Merger Rate

The major-merger rate per galaxy had conventionally been measured by two independent methods. J. M. Lotz et al. (2011) count morphologically disturbed systems using non-parametric indicators including the Gini coefficient, M_{20} (the second-order moment of light at the 20th-percentile brightness), CAS (Concentration, Asymmetry, Smoothness), and multi-mode/intensity/deviation $\{M, I, D\}$ statistics. They convert disturbed fractions into rates using a simulation-calibrated visibility timescale τ_{vis} , the time during which a major merger remains morphologically disturbed¹. Applied to close-pair counts in mass-selected SDSS samples, the morphological method gives a power-law redshift evolution

$$R_{\text{major}}(z) = R_0 (1 + z)^\alpha, \quad (1)$$

with $R_0 = 0.03 \pm 0.01 \text{ Gyr}^{-1}$ and $\alpha = +1.7 \pm 1.3$ for the J. M. Lotz et al. (2011) close-pair fit. V. Rodriguez-Gomez et al. (2015) measure the same rate by counting subhalo mergers in the Illustris cosmological hydrodynamical simulation, yielding $\alpha = +2.6 \pm 0.2$ for the Milky-Way-mass. At $z \approx 0.07$ this gives $R_{\text{major}} \approx 0.03\text{--}0.04 \text{ Gyr}^{-1}$, the local-universe reference value we compare against in Section 6.3.

The two $z \approx 0$ normalisations agree to within $\sim 1.5\times$, but the redshift-evolution exponents differ by roughly a factor of two. Both methods are indirect, with the morphological method converting a

disturbed-galaxy fraction into a rate through τ_{vis} , and the simulation method reading merger events off a subhalo tree of which realised mapping onto observable galaxies is itself not well understood.

2.3. The Galaxy Zoo 2 Catalogue

A complementary, more direct approach would be to instead ask classifiers, human or machine, whether a galaxy visibly looks disturbed or merging, and aggregate the resulting fractions across a flux- and resolution-limited sample.

2.3.1. Sample selection and imaging

GZ2 (K. W. Willett et al. 2013) solicits classifications from $\sim 84,000$ volunteer classifiers over ~ 14 months, totalling ~ 16 million votes on 304,122 SDSS DR7 Legacy galaxies selected with Petrosian magnitude $m_r < 17.0$, Petrosian half-light radius $r_{\text{Petro},90} > 3''$, and redshift $0.0005 < z < 0.25$ from the SDSS spectroscopic Main Galaxy Sample. The selection imposes the same flux and apparent-size cuts as GZ1 but is restricted to the brightest, best-resolved $\sim 34\%$ of the GZ1 sample so that deeper morphological questions (bars, arm count, dust lanes) can be more unambiguously identified.

Each galaxy is presented as an SDSS *gri* colour-composite JPEG cutout, scaled per-galaxy at $0.024 r_{\text{Petro},90}$ arcseconds per pixel and rendered at a fixed 424×424 pixels. The pixel scale therefore tracks each galaxy's physical extent rather than its on-sky size with angular footprints varying from cutout to cutout (roughly $10''$ for the smallest objects up to $\sim 2'$ for the largest), while the physical footprint is approximately $\sim 1.7 r_{\text{Petro},90}$ on a side, corresponding to $\sim 12 \text{ kpc}$ at the median GZ2 redshift $z \approx 0.07$.

2.3.2. Classification scheme

Classifiers answer an 11-question decision tree (Figure 2): Q1 (smooth/features/star), Q2 (edge-on?), Q3 (bar?), Q4 (spiral?), Q5 (bulge prominence), Q6 (anything odd?), Q7 (how round?), Q8 (odd feature: ring, lens/arc, disturbed, irregular, dust lane, merger, other), Q9 (edge-on bulge shape), Q10 (arm tightness), and Q11 (arm count). Each question is gated by a previous answer. For example, Q7 is asked only when Q1 = smooth, Q9 only when Q2 = yes, and so on. The 37 leaf answers form a hierarchical tree rather than a flat vector, and the reported labels are the fraction of classifiers who selected each leaf.

This tree expands substantially on the flat six-category GZ1 scheme of C. J. Lintott et al. (2008) (elliptical, clockwise spiral, anticlockwise spiral, edge-on disk, star/don't-know, merger). GZ2 drops the spiral chirality split, which proved susceptible to a documented left–right perception bias in GZ1 (K. Land et al. 2008), and adds eight new questions (Q2, Q3, Q5, Q7, Q8–Q11) for a sixfold increase in morphological resolution (K. W. Willett et al. 2013).

¹J. M. Lotz et al. (2008) calibrate τ_{vis} against hydrodynamic merger simulations and find $\tau_{\text{vis}} \approx 0.2\text{--}1.1 \text{ Gyr}$ depending on orbital parameters, mass ratio, and gas fraction.

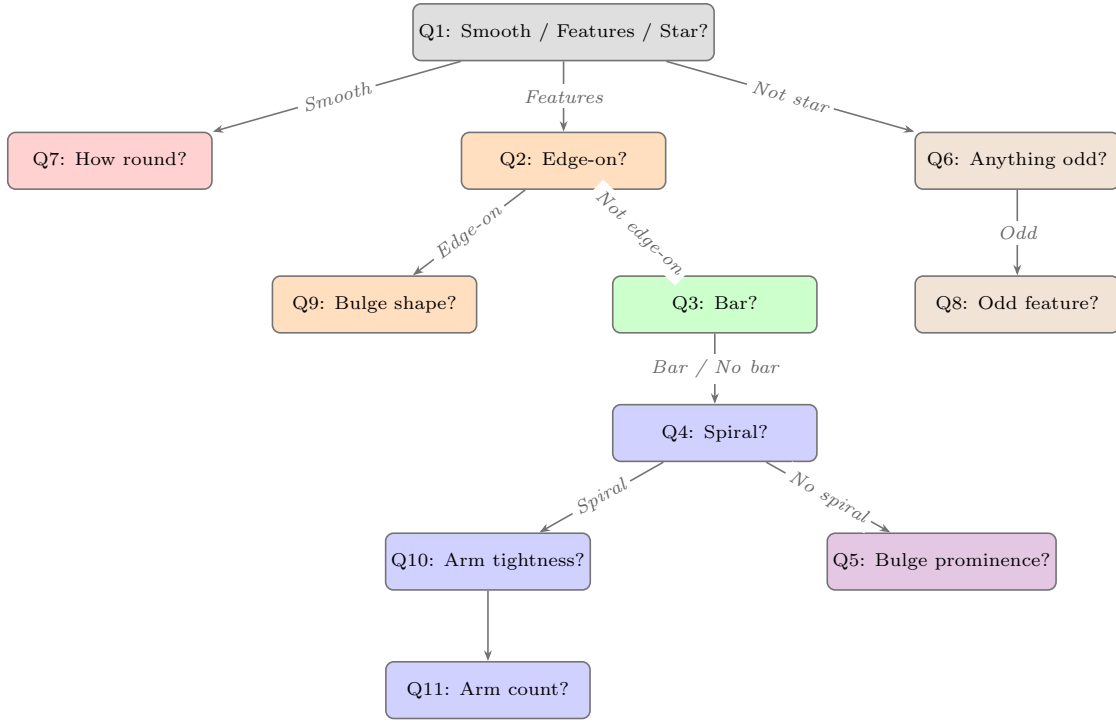


Fig. 2.— GZ2 hierarchical decision tree. Each node is one of the 11 GZ2 questions; directed edges show which downstream questions are gated by each answer at the parent node (e.g. Q7 “How round?” is only asked when a classifier first chose “Smooth” at Q1). The 37 leaf answers, the vote fractions we regress against, sit on the leaves of this tree (not shown for brevity). Node colours denote branch: gray = root, red = smooth, orange = edge-on, green = face-on/bar, blue = spiral, violet = bulge, brown = odd-feature.

2.3.3. Vote aggregation and cleaning

A typical galaxy receives 40–50 independent classifications, with a minimum of ~ 15 . Galaxies are shown to classifiers in randomised order, drawn from a queue that prioritises objects with the fewest existing classifications. To handle classifier noise and resolution-driven bias, K. W. Willett et al. (2013) apply three cleaning steps: (i) classifier-consistency weighting, in which each user’s vote weight is iteratively downweighted by their disagreement with the running consensus on a calibration sub-sample, suppressing the contribution of careless or adversarial classifiers; (ii) debiasing of vote fractions against apparent magnitude M_r and angular radius R_{50} following the method of S. P. Bamford et al. (2009); and (iii) parent-normalisation of child vote fractions to enforce the parent-sum consistency of the decision tree.

Together, the catalogue reports, for each question q , the classifier-consistency-weighted and resolution-debiased fraction $f_q \in [0, 1]$ of classifiers who selected that answer (we measure $\sim 55,000$ distinct values across our 61,578 training labels for Class1.1). Sibling fractions within a question sum to the parent’s vote fraction by construction. By reporting fractions rather than hard majority labels, we preserve classifier-disagreement since an ambiguous galaxy at $f_q \approx 0.5$ would be distinguishable from a confident one at $f_q \approx 1.0$.

2.3.4. Classification bias and apparent redshift evolution

For a galaxy with some morphology, at higher redshifts, it subtends a smaller angle, has lower surface brightness for resolved features, and is broadened by the SDSS $\sim 1.4''$ point-spread function. Spiral arms, bars, and bulge–disk features that are easily identified in a nearby galaxy are less resolved at $z \gtrsim 0.1$, so classifiers preferentially answer “smooth” and “round” for distant objects, surfacing as an apparent redshift evolution in mean GZ2 morphological type (S. P. Bamford et al. 2009).

K. W. Willett et al. (2013) correct for this following S. P. Bamford et al. (2009) by binning the sample in narrow redshift slices (where the intrinsic morphology mix is assumed fixed) and measuring how f_{feature} , f_{bar} , etc. depend on M_r and R_{50} within each slice; every galaxy’s raw vote fraction is then projected onto a well-resolved reference baseline ($M_r \approx -20.17$, $R_{50} \approx 3''$) at which the bias is by definition zero. Uniquely, the merger fraction f_{merger} cannot be recovered through this since at higher redshift, neighbouring galaxies blend into the target’s PSF and tidal features become harder to resolve, biasing the merger vote fraction upward rather than downward, so high- z galaxies are preferentially classified as mergers via the same resolution effect that pushes other features toward “smooth”. Therefore, while we use the corrected vote fractions as our labels, a residual resolution-dependent bias remains in the merger-

probabilities because the intrinsic rate also scales as $(1+z)^\alpha$ (J. M. Lotz et al. 2011; V. Rodriguez-Gomez et al. 2015).

2.3.5. Comparison to other morphological catalogues

Compared to alternative morphological catalogues, GZ2 has a larger sample size, containing $\sim 304,122$ galaxies, $\sim 20\times$ the P. B. Nair & R. G. Abraham (2010) (NA10) expert catalogue. Per-galaxy uncertainty is quantified by the ~ 40 independent votes rather than a single non-parametric scalar such as the de Vaucouleurs T-type (running from -6 for ellipticals to $+10$ for irregulars) or a Gini- M_{20} value. K. W. Willett et al. (2013) show that GZ2 vote fractions correlate strongly with NA10 T-types, so the citizen-science labels carry the same morphological information as expert classifications at a scale expert pipelines cannot match.

However, the drawback of any visual identification approach, be it expert T-typing, citizen-science voting, is that it returns a subjective answer where a non-parametric statistic such as G , M_{20} , CAS, or $\{M, I, D\}$ returns a deterministic scalar. The same pixels run through the Lotz pipeline produce the same Gini coefficient every time; the same pixels presented to two GZ2 classifiers, or to the same classifier on two different days, can produce different votes. Three galaxies from our own classification session on the GZ2 interface (<https://www.zooniverse.org/projects/zookeeper/galaxy-zoo/>) demonstrates this subjectivity (Figure 3).

The middle and bottom panels are more unambiguous, depicting a centrally concentrated smooth galaxy and a resolved spiral (2 spiral arms), on which essentially any classifier, human or non-parametric, could mostly return the same label. In contrast, the top panel depicts a bright bulge-like core embedded in low-surface-brightness extended emission for which the “spiral / not spiral” and “edge-on disk with dust lane” identifications could be visually degenerate at SDSS resolution, and the resulting vote fraction would be split. A non-parametric estimator would instead collapse such a galaxy to a deterministic scalar, while instead discarding the multi-feature information (e.g., bar presence, arm count, bulge prominence, oddity type) that GZ2 resolves across 37 separate leaves. A CNN trained on GZ2 would thus inherit both sides, where it operates at GZ2’s full 37-axis resolution, while propagating the per-image subjective noise that human classifiers contributed to the labels.

3. Data

The training set consists of 61,578 SDSS *gri* composite images (424×424 pixels) and 37 GZ2 classification labels per galaxy. Each label is a vote fraction in $[0, 1]$, the proportion of human classifiers who assigned the corresponding morphological feature. The labels are organized hierarchically: top-level questions (smooth vs. disk vs. star/artifact) gate whether deeper questions (bar, spiral, arm count) are asked.

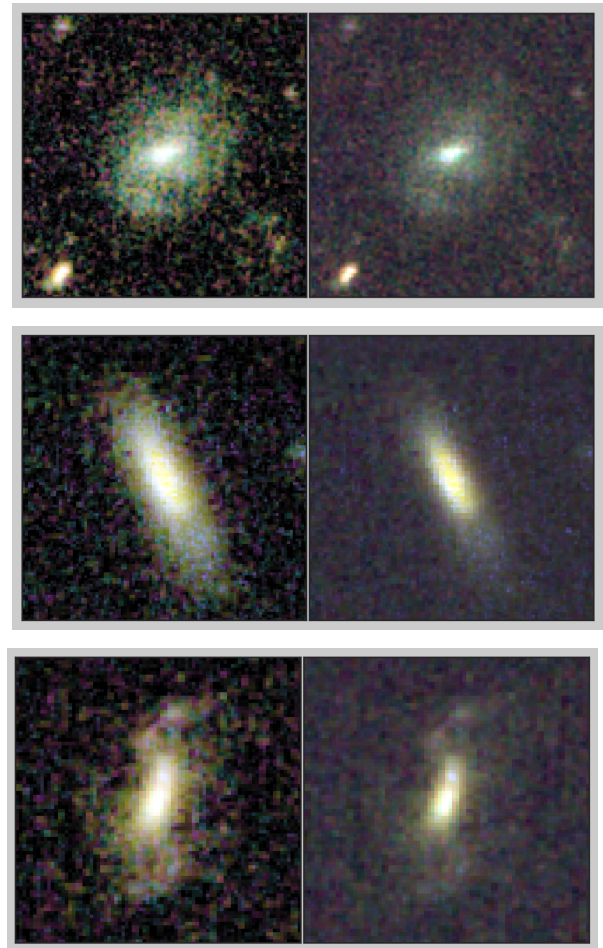


Fig. 3.— Three GZ2 galaxies from our own classification session. Top: an ambiguous case with a bright bulge-like core embedded in low-surface-brightness extended emission, where the Q4 “spiral / not spiral” and Q8 “dust lane” axes are visually degenerate at SDSS resolution, so the GZ2 vote fraction sits at intermediate values. Middle: a clean “smooth” galaxy with centrally concentrated mass, red, no resolved features. Bottom: a less-clean but visually discernable spiral galaxy with a clearly resolved disk with visible arm structure.

3.1. SDSS *gri* Composite Images

We use SDSS DR7 (K. N. Abazajian et al. 2009) three-colour cutouts in the *gri* bands, rendered as RGB JPEGs via the R. Lupton et al. (2004) scaling: *g* ($\lambda_{\text{eff}} \approx 477$ nm) to blue, *r* ($\lambda_{\text{eff}} \approx 623$ nm) to green, and *i* ($\lambda_{\text{eff}} \approx 763$ nm) to red. This mapping reproduces the colour-magnitude bimodality between blue-cloud star-forming disks ($g-r \sim 0.4$) and red-sequence quiescent ellipticals ($g-r \sim 0.8$ at $z \sim 0$). The CNN can therefore exploit pixel colour as a shortcut for the top-level Smooth/Features distinction; the harder labels (bar, arm count, dust lane, odd features) require genuine morphological structure beyond the dominant colour gradient.

The GZ2 sample spans $z \sim 0.02-0.25$ with median $z \approx 0.07$, drawn from the SDSS spectroscopic main galaxy sample (M. A. Strauss et al. 2002; C.

Stoughton et al. 2002). The native SDSS pixel scale is $0.396''/\text{pixel}$, corresponding to $\sim 0.54 \text{ kpc/pixel}$ at the median redshift assuming $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The GZ2 cutouts are pre-scaled per galaxy so that the field of view tracks $\sim 1.7 r_{\text{Petro},90}$ on a side, decoupling the rendered pixel scale from redshift (Section 3.6).

3.2. Label Distributions

Figure 4 shows normalized histograms of all 37 labels. The distributions fall into three families:

- **Top-level labels** (Class1.1–1.3, Class6.1/6.2): broad, roughly bimodal with most mass at 0 or 1, reflecting strong classifier consensus on the smooth-vs-disk distinction. The bimodality mirrors the red-sequence/blue-cloud dichotomy: pressure-supported quiescent ellipticals on one side, rotation-supported star-forming disks on the other.
- **Conditional sub-class labels** (Class7.x, Class10.x, Class11.x): concentrated near zero because most galaxies were never asked the corresponding gated question.
- **Rare-feature labels** (Class1.3 star/artefact, Class8.x odd features): sharply peaked at zero with a short tail, reflecting genuinely rare astrophysical states (rings, lens/arcs, dust lanes, mergers).

Two labels merit closer inspection. The merger vote (Class8.6) is heavily zero-concentrated because the visibly-disturbed merger phase lasts only ~ 0.5 – 1 Gyr against a quiescent lifetime of $\sim 10 \text{ Gyr}$, giving ~ 1 – 5% of galaxies in a merger state at any moment, consistent with J. M. Lotz et al. (2011). The cigar-shaped smooth label (Class7.3) is anomalous: its prototype images and correlations (Section 3.4) show that it predominantly picks up misclassified edge-on disks rather than true prolate spheroids, a confusion the CNN inherits from the training labels.

3.3. Prototype Images

Figure 5 shows the prototype image (highest vote fraction) for each label. Visually obvious labels (smooth, edge-on disk, grand-design 2-arm spiral) have clean, morphologically distinctive prototypes: featureless red ellipticals, thin elongated dust-lane-bisected disks, and clean logarithmic arms with blue star-forming knots tracing the density-wave crests. We expect the CNN to learn these labels well from a small number of templates.

Subtle or rare labels (lens/arc Class8.2, dust lane Class8.7, high-arm-count spirals Class11.3–11.5) require detecting faint, narrow, or rare features that are easily confused with PSF artefacts or flocculent disk structure. Merger prototypes (Class8.6) are visually heterogeneous, mixing interacting pairs, post-coalescence shells, and asymmetric isophotes across a wide range of merger stages and mass ratios. The

CNN therefore cannot learn a single “merger template”, a prediction borne out by the per-label RMSE analysis of Section 5.

3.4. Label Correlations

We compute the 37×37 Pearson correlation matrix (Figure 6) using

$$\rho_{ij} = \frac{\langle ij \rangle - \langle i \rangle \langle j \rangle}{\sqrt{\langle i^2 \rangle - \langle i \rangle^2} \sqrt{\langle j^2 \rangle - \langle j \rangle^2}}. \quad (2)$$

For a k -way question with perfect classifier consensus, sibling answers have expected correlation $\rho = -1/(k-1)$. For the effectively-binary Q1 (Class1.1 vs Class1.2; Class1.3 is near-zero everywhere, so Q1 collapses to a two-way vote in practice) and Q6 (Class6.1 vs Class6.2) we measure $\rho = -0.992$ and $\rho = -1.000$, matching the binary expectation of -1 . For the $k = 4$ bulge-prominence question Q5 we measure -0.20 to -0.40 , near the expected $-1/3$, with departures consistent with deeper-tree classifier noise. This confirms that the W13 debiasing preserves within-question exclusivity.

The strongest parent–child correlations are Edge-on Yes $\rightarrow$ Rounded bulge ($\rho = +0.879$), Spiral Yes $\rightarrow$ Medium winding ($\rho = +0.858$), Spiral Yes $\rightarrow$ 2 arms ($\rho = +0.781$), and Features/Disk $\rightarrow$ Edge-on No ($\rho = +0.748$). The last is asymmetric: Features/Disk correlates with Edge-on No at $+0.748$ but with Edge-on Yes at only $+0.333$. The asymmetry follows from the $\cos i$ inclination distribution of randomly-oriented thin disks: truly edge-on views ($i > 80^\circ$) subtend only $\sim 17\%$ of the solid angle, so most disk galaxies are face-on or moderately inclined.

The Cigar-shaped label (Class7.3) is anomalous. Nominally a smooth sub-label, it correlates positively with Edge-on Yes ($\rho = +0.496$) and with the edge-on bulge labels Class9.1 ($\rho = +0.370$) and Class9.3 ($\rho = +0.475$), but is essentially uncorrelated with Smooth itself ($\rho = -0.001$). Highly inclined disks are therefore systematically misclassified as “smooth + cigar-shaped” when the disk substructure is unresolved at SDSS seeing, a confusion the CNN inherits from the labels.

3.5. Diagnostic Analyses

We perform four diagnostic analyses of the label space. Figure 2 visualizes the GZ2 hierarchical decision tree and the gating structure between questions. Figure 7 reorders the correlation matrix by tree branch, exposing block structure that the unordered matrix obscures. Figure 8 shows a t-SNE embedding of the 37-dimensional label space: the manifold separates cleanly along the smooth/disk axis but rare classes form sparse, isolated clusters. Figure 9 plots the effective sample size

$$N_{\text{eff}} = \frac{(\sum_i f_i)^2}{\sum_i f_i^2}, \quad (3)$$

per label, with f_i the corrected vote fraction for galaxy i . N_{eff} counts effective contributors: a label that is

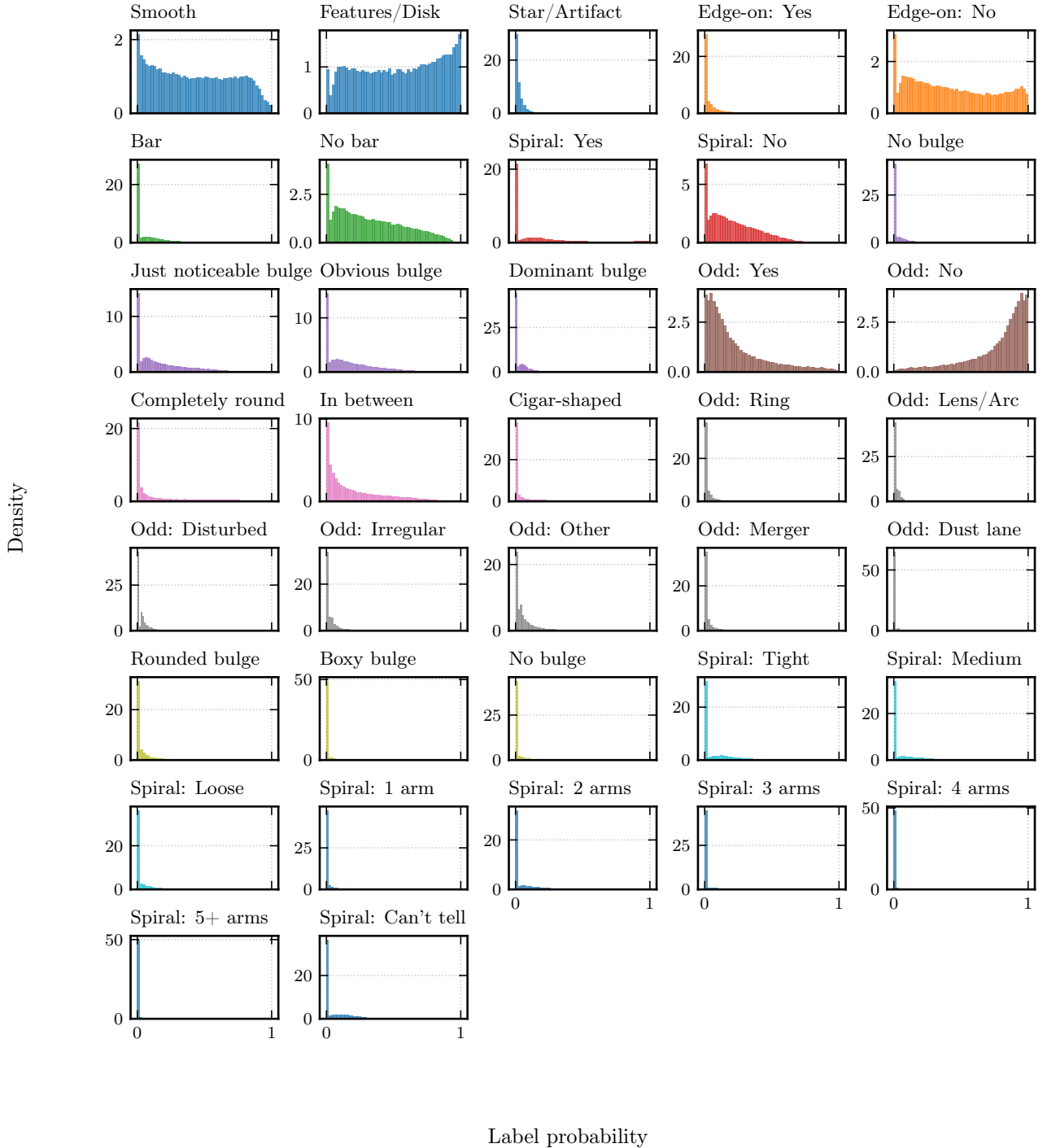


Fig. 4.— Normalized histograms of all 37 GZ2 classification labels. Top-level labels show broad distributions; deeper hierarchical labels are concentrated near zero. The top-level vote-fraction labels (Class1.1/1.2, Class6.1/6.2) display broad bimodal distributions with mass piled at 0 and 1, while conditional sub-class labels (Class7.x, Class10.x, Class11.x) are sharply concentrated near zero because most galaxies were never asked the corresponding gated question.

$f = 1$ on N galaxies and 0 elsewhere has $N_{\text{eff}} = N$, while a label concentrated on a single galaxy has $N_{\text{eff}} = 1$. Deeper-tree labels reach $N_{\text{eff}} \lesssim 1000$, far below the nominal 61,578.

Figure 10 shows that N_{eff} and the mean-prediction baseline RMSE are strongly correlated across labels. Rare labels (lens/arc, dust lane, high-arm-count spi-

als) achieve low baseline RMSE because predicting zero is already accurate when the label is near-zero everywhere, not because the model has learnt them. Per-label RMSE numbers in Section 5 are therefore only informative for the well-sampled labels in the upper-right of the figure.

Effective dimensionality of the label space.

Figure 11 shows the eigenspectrum of the 37×37 label correlation matrix and the cumulative variance fraction. The Kaiser criterion ($\lambda > 1$) gives an effective dimensionality of ~ 8 , far fewer than the nominal 37, and $\sim 90\%$ of the variance is captured by the first ~ 10 principal components. The CNN is therefore not solving 37 independent regression problems; the effective task complexity reduces to a handful of latent morphological factors. We take this as an upper bound on the representational capacity the FC head needs, and size the FC-head parameter count accordingly in Section 4.3.

Geometry of the vote-fraction distributions.

Figure 12 shows pairwise scatter plots for the eight most strongly correlated label pairs (four positive, four negative). Positively-correlated pairs lie along a diagonal with most galaxies stacked at the (0,0) and (1,1) corners; negatively-correlated pairs form L-shapes confined to the high- i /low- j and low- i /high- j arms. Vote fractions saturate at 0 or 1 when classifier consensus is strong, and the interior of $[0,1]^2$ is sparsely populated. This corner-saturating density motivates the sigmoid output head used by both architectures in Section 4.1, which naturally saturates at 0 and 1 to match the label density.

Figure 13 shows image montages from the four quadrants of the (Features/Disk, Edge-on-No) plane. The high-high and low-low corners contain archetypal disks and ellipticals respectively, while the off-diagonal quadrants are populated by inclined disks and ambiguous low-surface-brightness galaxies. The L-shape structure therefore reflects visually distinct astrophysical classes rather than decision-tree gating alone.

Figure 14 drills into the positively-correlated Spiral Yes / Medium-winding pair. Galaxies with high vote fractions on both labels are intermediate-winding (S_b -type) spirals; the off-diagonal quadrants pick up smooth ellipticals or tightly-wound grand-design spirals whose votes shift weight to the tight-winding sibling. The four quadrants of every positively-correlated pair separate into morphologically meaningful sub-populations, confirming that the strong off-diagonal correlations of Figure 6 reflect real morphological structure rather than residual gating artefacts.

Colour and brightness shortcut for top-level labels. Smooth ellipticals are redder, more centrally concentrated, and higher-contrast than disk galaxies at the pixel level (Section 3.1). The CNN can therefore answer the Q1 Smooth-vs-Features distinction from global pixel statistics (centre-weighted brightness, red-to-blue ratio, intensity dispersion) without recognising any morphological feature. The bulk-RMSE wins in Section 5 draw on this shortcut; the harder labels (bar, arm count, dust lane, merger) cannot, and their high per-label RMSE reflects the absence of the colour shortcut.

3.6. Image Preprocessing

Raw GZ2 cutouts are $424 \times 424 \times 3$ in *gri* RGB, most of which is empty sky. At full resolution the 61,578-galaxy training set requires ≈ 133 GB as `float32`, well above the 12 GB VRAM of our local NVIDIA RTX 5070 Ti laptop GPU. The pipeline therefore:

- Centre-crops by 15% per side (retaining the central $\sim 296^2$ region around the galaxy).
- Bilinearly resizes to a 136×136 on-disk cache in `uint8` (~ 3.4 GB total).
- GPU-side **CenterCrop(96)** at train time yields the $96 \times 96 \times 3$ model input.

The 96×96 input is chosen for image quality rather than memory. Table 1 lists the nested pixel scales: the native SDSS scale of $0.396''/\text{pixel}$, the per-galaxy-scaled GZ2 cutout at $\sim 0.020''/\text{pixel}$ (≈ 27 pc), and the 96×96 model input at $\sim 0.040''/\text{pixel}$ (~ 60 pc) spanning ~ 5.7 kpc on a side at $z \approx 0.07$.

Stage	Angular	Physical (at $z = 0.07$)
Native SDSS imaging	$0.396''/\text{pix}$	~ 540 pc/pix
GZ2 per-galaxy cutout (424^2)	$\sim 0.020''/\text{pix}$	~ 27 pc/pix
Model input (96^2)	$\sim 0.040''/\text{pix}$	~ 60 pc/pix
SDSS PSF FWHM	$\sim 1.4''$	~ 1.9 kpc

Table 1: Nested pixel scales of the pipeline at the median GZ2 redshift $z \approx 0.07$. The model input field of view spans ~ 6 kpc on a side. The SDSS point-spread-function FWHM is the fundamental resolution limit and is several model pixels wide, we are PSF-limited, not pixel-limited, for sub-kpc features.

The morphological feature scales of interest fit within the 96^2 field of view: spiral arm widths (~ 0.5 – 2 kpc; C. J. Conselice 2014) span ~ 10 – 30 pixels, bar lengths (~ 2 – 8 kpc) span ~ 30 – 100 pixels, bulge effective radii ($r_e \sim 0.5$ – 2 kpc) span ~ 10 – 30 pixels, and edge-on dust lanes ($h_z \sim 100$ – 300 pc) span ~ 2 – 5 pixels at the resolvability limit. The SDSS PSF ($\sim 1.4''$ FWHM, ~ 1.9 kpc at $z \approx 0.07$) sets the fundamental resolution limit, so 96^2 is the coarsest scale that still resolves all five categories at the median redshift. We adopt 96×96 rather than the lab manual’s suggested ~ 64 because at ~ 60 pc/pixel we still resolve the dust-lane scale; going coarser would merge the dust-lane and bulge categories at the median redshift.

Cache size. The 136×136 cache is sized by the rotation-augmentation constraint of Section 4.4: an inscribed square of side s inside an outer square of side S at 45° rotation requires $S \geq s\sqrt{2}$, so $s = 96$ implies $S \geq \lceil 96\sqrt{2} \rceil = 136$. Caching at 136^2 and centre-cropping to 96^2 after rotation guarantees that no black corner pixels ever enter the network at any rotation angle. We therefore set the pre-crop fraction to 0.15 rather than the more obvious 0.25: the rotation-safe-buffer derivation fixes the value, not visual framing.

3.7. Train/Validation Split

We randomly split the dataset 80/20 (seed 42) into 49,262 training and 12,316 validation images. Figure 16 confirms that the per-label distributions are statistically indistinguishable between the two sets; per-label two-sample Kolmogorov–Smirnov tests do not reject the null hypothesis of identical distributions at $\alpha = 0.05$ for any label.

3.8. Baseline Model

We minimise the root-mean-square error between predicted and true vote-fraction matrices,

$$L_{\text{RMSE}} = \sqrt{\frac{1}{N_{\text{gal}} N_{\text{lab}}} \sum_{i=1}^{N_{\text{gal}}} \sum_{j=1}^{N_{\text{lab}}} (\ell_{ij}^{\text{true}} - \ell_{ij}^{\text{pred}})^2}, \quad (4)$$

where $\ell_{ij}^{\text{true}}, \ell_{ij}^{\text{pred}} \in [0, 1]$ are the true and predicted vote fractions for galaxy i and label j , and $N_{\text{lab}} = 37$. RMSE rather than BCE because the task is regression on continuous vote fractions, not classification, and the resulting scalar shares units with the labels themselves so a value of 0.10 is directly interpretable as “ten percentage points off on average.”

The baseline model predicts the training-set mean label vector for every image, yielding training and validation RMSEs of 0.1638 and 0.1639 respectively. Any CNN must outperform this floor.

4. Methods

4.1. Network Architectures

We train two backbone architectures: a hand-designed 4-block Custom CNN ($\sim 2.8\text{M}$ parameters) and an 18-layer ResNet ($\sim 11.2\text{M}$ parameters). Both consume $96 \times 96 \times 3$ inputs and emit a 37-dimensional sigmoid head, constraining each output to $[0, 1]$ to match the vote-fraction range. They differ in depth, parameter count, and the use of residual connections. Figure 17 summarizes the two layer stacks side-by-side.

Layer primitives. A 2D convolution with kernel k of size $K \times K$ maps an input feature map x of shape (C_{in}, H, W) to an output of shape (C_{out}, H, W) via

$$y_{c,i,j} = \sum_{c'=1}^{C_{\text{in}}} \sum_{u=0}^{K-1} \sum_{v=0}^{K-1} k_{c,c',u,v} x_{c',i+u-\lfloor K/2 \rfloor, j+v-\lfloor K/2 \rfloor} + b_c, \quad (5)$$

with reflection padding by $\lfloor K/2 \rfloor$ preserving the spatial extent. The learned tensor k has $C_{\text{out}} C_{\text{in}} K^2$ weights and C_{out} biases per layer; spatial weight sharing is what makes a CNN cheaper than the equivalent fully-connected layer by a factor of HW . Each conv is followed elementwise by a rectified linear activation

$$\text{ReLU}(z) = \max(0, z), \quad (6)$$

and the final pre-output activation is the sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad (7)$$

which saturates to the unit interval and matches the corner-saturating geometry of the vote-fraction distributions (Section 3.5). Spatial downsampling is performed by max pooling over non-overlapping 2×2 windows,

$$y_{c,i,j} = \max_{(u,v) \in [0,1]^2} x_{c,2i+u,2j+v}, \quad (8)$$

which halves H and W at each application while preserving the brightest activation in each window so that locally bright morphological features (arm crests, bar ends, bulge cores) survive the downsampling.

4.2. ResNet-18

Residual networks. Plain deep CNNs suffer from a degradation problem: as more layers are stacked, training accuracy saturates and then degrades, driven by the vanishing-gradient problem. The gradient of the loss with respect to an early-layer weight is a product of N Jacobians whose operator norms are typically < 1 , so $\|\nabla_{W_\ell} L\| \sim \prod_{k=\ell+1}^N \|J_k\|$ decays geometrically with depth. K. He et al. (2015) reformulate each block to learn the residual $F(x) = H(x) - x$ rather than the full mapping $H(x)$, adding an identity shortcut so that

$$H(x) = F(x) + x, \quad J_{\text{block}} = I + \frac{\partial F}{\partial x}. \quad (9)$$

Gradients then propagate through the skip unattenuated since the additive identity in J_{block} guarantees $\|J_{\text{block}}\| \geq 1$ in the small-perturbation limit; extra layers can always learn $F \equiv 0$ and recover a shallower network, so depth cannot hurt training error. K. He et al. (2015) use this property to train networks up to 1202 layers on CIFAR-10.

ResNet-18 adaptation. We use the standard `torchvision` ResNet-18 stem (a 7×7 stride-2 convolution followed by 3×3 stride-2 max-pool) without modification. The 96×96 input downsamples to 24×24 before stage 1, and the four stride-2 reductions inside the residual stages take the spatial dimensions to 3×3 at the global-average-pool layer. We replace the classification head with a single linear layer mapping the 512-dimensional feature vector to 37 sigmoid-activated outputs. The total parameter count is **11,195,493**. We train with the Adam optimiser (fused, initial LR $\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$) and the RMSE loss for 100 epochs at batch size 1024, with no augmentation or LR scheduler in this baseline; those interventions are added in Section 4.4. Adam updates each parameter θ from gradient $g_t = \nabla_\theta L_t$ via bias-corrected first and second moment estimates,

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad \theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{v_t}} m_t \quad (10)$$

with $\hat{m}_t = m_t / (1 - \beta_1^t)$ and $\hat{v}_t = v_t / (1 - \beta_2^t)$; the per-parameter step size $\eta / \sqrt{\hat{v}_t}$ adapts to gradient magnitude, which lets the same global LR drive both the small-scale conv weights and the large-scale FC head without manual rescaling (D. P. Kingma & J. Ba 2014).

Per-stage physical scales. At the median GZ2 redshift, ~ 60 pc per model pixel, the four ResNet stages operate on distinct physical regimes:

- Stem ($96 \rightarrow 24, 64$ ch): ~ 240 pc/pixel low-pass response.
- Stage 1 ($24 \rightarrow 24$): texture and local contrast (smooth vs. clumpy light, disk–bulge transition).
- Stage 2 ($24 \rightarrow 12, 128$ ch): arms, bars, and bulges at ~ 480 pc/pixel (Q4/Q5/Q10/Q11).
- Stage 3 ($12 \rightarrow 6, 256$ ch): galaxy-scale gestalt at ~ 1 kpc/pixel (Q2 edge-on, Q6/Q8 asymmetry).
- Stage 4 ($6 \rightarrow 3, 512$ ch): whole-galaxy fingerprint at ~ 2 kpc/pixel.

The skip connections preserve fine-scale features from earlier stages into the global feature vector, which is particularly important for rare-feature labels (lens/arc, dust lane).

4.3. Custom CNN

Architecture. The Custom CNN stacks four convolutional blocks of $\{\text{Conv2d}(3 \times 3, \text{pad } 1) \rightarrow \text{BatchNorm2d} \rightarrow 3 \times 3 \text{ kernels are standard at this spatial size. The sigmoid output matches the } [0, 1] \text{ vote-fraction range; the loss is RMSE rather than BCE because the task is regression on continuous vote fractions, not classification.}$ with channel widths $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$. The spatial size halves at each block, so the 96^2 input collapses to a $6^2 \times 256$ feature map. The classifier head is two fully connected layers ($9216 \rightarrow 256 \rightarrow 128$) with ReLU activations and tapered dropout (0.25 after FC1, 0.10 after FC2), followed by a linear-plus-sigmoid layer to the 37 vote fractions. We train with Adam at initial LR 10^{-3} for 100 epochs at batch size 1024.

Batch normalisation and dropout. Batch normalisation rescales each channel of a feature map to zero mean and unit variance over a mini-batch before applying a learned affine,

$$\hat{x}_c = \frac{x_c - \mu_{B,c}}{\sqrt{\sigma_{B,c}^2 + \epsilon}}, \quad y_c = \gamma_c \hat{x}_c + \beta_c, \quad (11)$$

with batch statistics $\mu_{B,c}, \sigma_{B,c}^2$ replaced by running estimates at inference (S. Ioffe & C. Szegedy 2015). BN keeps the input distribution of each conv layer stationary across training, allowing the larger learning rate (10^{-3}) and shorter warm-up than a BN-free stack of the same depth would tolerate, and contributes a mild stochastic regularisation through the noise in the per-batch statistics. Dropout, applied to the FC head, sets each unit to zero independently with probability p at training time and rescales the survivors,

$$y_i = \frac{1}{1-p} x_i r_i, \quad r_i \sim \text{Bernoulli}(1-p), \quad (12)$$

collapsing to the identity at inference ($p = 0$) so no scaling correction is needed at test time (N. Srivastava et al. 2014). The heavier dropout on FC1 ($p = 0.25$) concentrates regularisation in the $9216 \rightarrow 256$ matrix, which carries $\sim 85\%$ of the FC-head parameters.

Depth and dropout ablation. We select the depth-4 / tapered-dropout configuration via an ablation sweep at 25 epochs on the same train/val split, varying depth $\in \{3, 4, 5\}$ at fixed tapered dropout and dropout $\in \{(0.10, 0.10), (0.25, 0.10), (0.25, 0.25)\}$ at fixed depth 4. At depth 3 the post-conv feature map is $12^2 \times 128 = 18,432$ values, inflating the FC head and increasing overfitting risk; at depth 5 the feature map collapses to $3^2 \times 512 = 4,608$, erasing spatial structure. Depth 4 is the balance point. Figure 19 shows that depths 3, 4, and 5 cluster within run-to-run noise, while the tapered-dropout pivot beats the uniform (0.25, 0.25) variant and tracks the uniform (0.10, 0.10) variant. We lock in the (depth-4, tapered) configuration for the 100-epoch run.

Design choices. The channel progression $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ tracks the same physical-scale hierarchy as the ResNet stages of Section 4.2 (block 1 at ~ 0.2 kpc/pixel picks up local isophote gradients; block 4 at ~ 1.8 kpc/pixel produces a galaxy-scale fingerprint). We use max pooling rather than average pooling so that locally bright features (arm crests, bar ends, bulge cores) survive the downsampling. The sigmoid output matches the $[0, 1]$ vote-fraction range; the loss is RMSE rather than BCE because the task is regression on continuous vote fractions, not classification.

Parameter count. The Custom CNN has **2,786,117** trainable parameters, of which ~ 2.36 M live in the $9216 \rightarrow 256$ FC1 matrix. The training set contains $N_{\text{train}} \times 96^2 \times 3 \approx 1.36 \times 10^9$ pixels, giving a parameter-to-pixel ratio of $\sim 2 \times 10^{-3}$ for the Custom CNN and $\sim 8 \times 10^{-3}$ for ResNet-18. Both are well below unity, but neighbouring pixels are highly correlated and most pixels are background sky, so the effective independent-sample count is much smaller. The train/val gaps in Figures 18 and 20 confirm modest overfitting, motivating the regularisation interventions of Section 4.4.

Model selection. The Custom CNN and ResNet-18 are indistinguishable within ~ 0.0005 RMSE in this unaugmented baseline (Custom: 0.0953, ResNet-18: 0.0958). The lab manual instructs us to “use the best performing model between your own implementation and ResNets” for the rest of the lab; we satisfy this by tie-breaking on parameter efficiency: the Custom CNN has $\sim 4\times$ fewer parameters, $\sim 2\times$ faster inference, and a plain Conv–BN–ReLU–Pool stack that isolates the marginal effects of the scheduler and augmentation. **All downstream experiments use the Custom CNN as the backbone.**

4.4. Optimization Interventions

We add two interventions on top of the Custom CNN baseline: an adaptive learning-rate scheduler

and random-angle rotation augmentation. The scheduler addresses optimisation (fine-tuning the descent in the existing loss landscape); the augmentation addresses generalisation (expanding the effective training distribution). The two are nearly orthogonal and are expected to stack.

4.4.1. Learning-rate scheduling

We use `torch.optim.lr_scheduler.ReduceLROnPlateau` with `factor = 0.5` and `patience = 3`. The scheduler halves the learning rate whenever the validation loss has not improved for 3 epochs, adapting to the actual convergence trajectory rather than a hand-tuned schedule.

Figure 21 shows the loss and learning-rate trajectories. The LR drops in factor-of-two steps each time the validation loss plateaus, taking it from 10^{-3} to $\sim 10^{-4}$ over ~ 3 –4 steps in the 50-epoch run. Each drop is mirrored by a small downward kink in the validation curve at the same epoch, confirming that the scheduler was breaking plateaus rather than minima. The scheduled Custom CNN reaches a validation RMSE of 0.0943 at epoch 40, ~ 0.001 below the unscheduled baseline (0.0953).

4.4.2. Rotation augmentation

Augmentation is only safe when the transformation is a label-symmetry; otherwise the network trains on examples with incorrect labels. We enumerate candidate transformations:

- **Rotation** by $\theta \in [0, 360^\circ]$: exact sky symmetry, label-preserving because the GZ2 corrected vote fractions do not encode chirality (the GZ1 spiral-winding-direction branch was dropped in GZ2; K. W. Willett et al. 2013). **Adopted.**
- **Translation, scaling, brightness/contrast/colour jitter, additive noise**: rejected. Each either moves the galaxy out of the field of view or simulates the resolution-, surface-brightness-, and magnitude-driven classification bias that the W13 debiasing removes.
- **Parity (reflection)**: label-preserving but redundant under $[0, 360^\circ]$ rotation.

We apply rotation per-batch on the GPU via `affine_grid + grid_sample` (bilinear). The $136 \rightarrow 96$ cache geometry guarantees the rotated image fully covers the 96×96 centre region at every angle.

We verify the augmentation in isolation with a 50-epoch quick test at fixed learning rate (Figure 22). Together with the unaugmented baseline (Figure 20) and the scheduler-only run (Figure 21), this completes a 2×2 ablation across the scheduler and augmentation axes. The augmentation-only run reaches a best validation RMSE of 0.0915 at epoch 48, the largest single-intervention gain ($\Delta\text{RMSE} \approx -0.0038$); the scheduler-only gain ($\Delta\text{RMSE} \approx -0.0010$) is $\sim 4\times$ smaller. Combining both (Figure 23) delivers the best overall RMSE of **0.0913**.

Figure 23 shows the final combined run. The train/val gap at each model’s best epoch quantifies the effect: the unaugmented Custom CNN at epoch 34 has train RMSE 0.0888 and val RMSE 0.0953 (gap $+0.0064$), while the augmented and scheduled model at epoch 97 has train RMSE 0.0929 and val RMSE 0.0913 (gap -0.0016). The gap has closed by a factor of ~ 4 and reversed sign: the augmented training set is harder than the unrotated validation set, so the trained model performs better on validation. The training curve no longer descends monotonically because every epoch presents a fresh random angle per image, preventing memorisation of specific (image, label) pairs. We take the persistent per-epoch wobble as the operational definition of “not memorising”: the network sees rotated views it has not seen before at every epoch, so the training loss never collapses onto the validation loss the way it does under deterministic input.

4.5. Tree-Reweighted Output (Extra Credit)

The 37 GZ2 labels are not independent: by survey construction, a child vote fraction f_q is the product of its conditional probability given that the classifier reached question q and its parent’s vote fraction,

$$f_q = p_{q|\text{parent}} \prod_{a \in \text{ancestors}(q)} p_{a|\text{parent}(a)}. \quad (13)$$

An independent-output model is free to predict $\hat{f}_{1.1} = 0.10$ and $\hat{f}_{7.1} = 0.50$, which is physically impossible since Class7.1 only exists when Q7 was asked and Q7 is only asked on the Class1.1 branch. Following Task 22 option (a), we sigmoid-activate the 37 raw outputs and interpret them as conditional probabilities $\hat{p}_{q|\text{parent}}$, then walk the GZ2 decision tree in topological order to produce tree-consistent vote-fraction predictions $\hat{f}_q$. For Class4.1 and Class4.2 (two possible parents Class3.1 or Class3.2) we generalise to $\hat{f}_q = \hat{p}_{q|\text{parent}} \sum_p \hat{f}_p$, since within a single GZ2 question the parents are mutually exclusive. The output is then guaranteed tree-consistent: $\hat{f}_{\text{child}} \leq \sum_p \hat{f}_p$ for every child–parent set.

The architectural change is purely in the output transformation; the loss remains RMSE, so the bulk validation RMSE is directly comparable to the augmented Custom CNN.

The tree-reweighted model attains a validation RMSE of 0.2066 versus 0.0913 for the augmented Custom CNN (Figure 24).

The constraint $\hat{f}_{\text{child}} \leq \sum_p \hat{f}_p$ prevents the network from placing probability mass on a deep-tree label when its parent is low. Table 2 shows that the labels most penalised are the rare deep-tree odd-feature labels (lens/arc, dust lane, merger). The independent-output model achieved low RMSE on these labels by predicting small positive values; the constraint now forces the prediction to zero exactly when the parent is also low. The deep-tree-label improvements promised by the constraint are real within the smooth and disk sub-trees, but they are washed out in bulk RMSE

by the strict enforcement on rare labels. The tree-reweighted model is the more physically principled architecture, but the unconstrained model is the better bulk-RMSE minimiser; we carry the augmented unconstrained model forward as the working best model.

Label	Indep. RMSE	Tree-RW Δ RMSE
Spiral: Loose	0.0799	+0.0309
Spiral: 2 arms	0.1377	+0.0161
Obvious bulge	0.1274	+0.0216
Star/Artefact	0.0344	+0.4420
Odd: Dust lane	0.0252	+0.2221
Odd: Lens/Arc	0.0271	+0.2111
Odd: Merger	0.0484	+0.1745
Edge-on: Yes	0.0864	+0.1741

Table 2: Per-label validation RMSE change (tree-reweighted minus independent) for selected labels. Deep-tree rare-feature labels (Star/Artefact, Lens/Arc, Dust lane, Merger) suffer the largest increase because the architectural constraint forces them to exactly zero when their parent is also low; the independent-output baseline was achieving low RMSE on these labels by predicting near-zero values rather than enforced-zero, which happens to score better in bulk RMSE.

5. Results

5.1. Validation RMSE and Model Comparison

Figure 25 compares the validation RMSE trajectories of all trained models; Table 3 summarises the best validation RMSE for each. The Custom CNN with rotation augmentation and `ReduceLROnPlateau` scheduling achieves the best performance: RMSE = 0.0913, a 44% reduction below the mean-prediction baseline (0.1639) and below the lab manual’s “good” target of 0.09. The tree-reweighted Custom CNN sits well above the others in bulk RMSE because the architectural constraint shrinks the prediction manifold; we carry the augmented unconstrained Custom CNN forward as the best model.

Model	Best val. RMSE	Best epoch
Mean-prediction baseline	0.1639	34
Custom CNN (plain)	0.0958	95
ResNet-18 (plain)	0.0943	40
Custom CNN + LR scheduler	0.0915	48
Custom CNN + augmentation	0.0913	97
Custom CNN + augmentation + LR scheduler	0.0913	97
Custom CNN + tree-reweighted output	0.2066	63

Table 3: Best validation RMSE achieved by each model. The tree-reweighted model’s high bulk RMSE is an artefact of the architectural constraint, not a generalisation failure, see Section 4.5 and Table 2.

The progression baseline $\rightarrow$ plain backbone $\rightarrow$ LR scheduler $\rightarrow$ augmentation lowers the validation

RMSE monotonically. The largest single gain comes from rotation augmentation (Δ RMSE = -0.0028 over the plain Custom CNN); the scheduler gain is Δ RMSE = -0.0010 . The combined-intervention performance approximately equals the sum of individual gains, consistent with the two interventions addressing orthogonal failure modes. Figure 26 shows the same information as a step-wise bar chart: the bulk of the reachable improvement over the baseline comes from the architectural step to a CNN backbone, with the optimisation interventions providing smaller cumulative refinements toward the “good” target.

5.2. Per-Label Performance

Figure 28 shows true-vs-predicted scatter plots for all 37 labels on the validation set using the best Custom CNN. The labels fall into three performance tiers, which map onto the rarity-vs-baseline-RMSE diagnostic of Figure 10:

- **Well-sampled top-level labels** (Class1.1 Smooth, Class1.2 Features/Disk, Class2.1 Edge-on, Class4.1 Spiral, Class6.x Odd): scatter plots track the 1:1 line cleanly across $[0, 1]$ with small bias and $\sigma \sim 0.07\text{--}0.12$. The CNN is genuinely learning morphology.
- **Moderate-sample mid-tree labels** (Class5.x bulge prominence, Class10.x arm tightness, Class11.2 two-arm spiral, Class7.x roundness): clear positive correlation with regression toward the mean. Bias remains small, scatter is larger ($\sigma \sim 0.10\text{--}0.15$).
- **Sparse deep-tree labels** (Class1.3 star/artefact, Class8.x odd features, Class11.3–11.5 high-arm-count): predictions collapse to a band near $\hat{f} = 0$. The model predicts near-zero everywhere, achieving low bulk RMSE without distinguishing positive examples.

Figure 27 summarises per-label RMSE quantitatively. The hardest labels (largest RMSE) are the bimodal top-level votes (smooth, features/disk, edge-on, obvious bulge), where any miscalibration is visible; the apparently easiest labels (lens/arc, dust lane, high-arm-count) are easy only because they are near-zero almost everywhere.

5.3. Extreme Examples per Label

Figures 29–35 show the top-5 most extreme validation images for each of seven prototype labels, with actual-label ordered and predicted-ordered rows for comparison. We summarise the seven cases:

- **Smooth** (Class1.1): actual and predicted top-5 essentially agree. Round, centrally concentrated, featureless ellipticals; the model is genuinely calibrated on this well-sampled, bimodal label.

- **Star/Artefact** (Class1.3): actual top-5 are point sources; predicted top-5 occasionally include compact-but-extended galaxies because the network defaults to compactness as a proxy when the class is sparse.
- **Edge-on disk** (Class2.1): strong agreement. Edge-on disks are visually distinctive and well-sampled.
- **Odd: Ring** (Class8.1): partial agreement. Predicted top-5 include face-on barred spirals where the network has latched onto inner-Lindblad-resonance and outer-Lindblad-resonance pseudorings.
- **Odd: Lens/Arc** (Class8.2): poor agreement. Lens/arc systems are vanishingly rare ($N_{\text{eff}} < 200$); the predicted top-5 mostly flag galaxies with faint diffuse outer envelopes as arc-like.
- **Spiral: 2 arms** (Class11.2): excellent agreement. Both rows are grand-design spirals with clear $m = 2$ patterns, one of the model’s cleanest visual templates.
- **Odd: Merger** (Class8.6): partial agreement. The predicted top-5 mix genuine mergers with asymmetric or low-surface-brightness systems flagged as “disturbed-looking”. This failure mode makes the merger-fraction estimate of Section 5.4 an upper bound and motivates the threshold-sweep there.

5.4. Merger Fraction on the Test Set

We apply the augmented and LR-scheduled Custom CNN to a held-out test set of 79,975 SDSS images and estimate the merger fraction as the mean predicted probability of the “Odd: Merger” label (Class8.6):

$$f_{\text{merger}} = 4.01\% \pm 0.03\% \text{ (stat, } 1\sigma \text{ bootstrap)}, \quad (14)$$

with a 95% bootstrap interval of 3.95–4.06%. The statistical uncertainty is small because the test sample is large; the systematic uncertainties are discussed below.

Threshold sensitivity. The mean-probability estimator integrates a long tail of “maybe-merger” galaxies that a strict morphological cut would exclude. Figure 36 plots $f_{\text{merger}}(p > p_{\text{cut}})$ as a function of the probability cut and overlays the ~ 0.5 –3% band expected at $z \approx 0$ from J. M. Lotz et al. (2011). Our curve enters the Lotz band at $p_{\text{cut}} \approx 0.3$ –0.5: requiring high-confidence merger predictions recovers the literature value almost exactly.

From fraction to rate. The Lotz major-merger rate is quoted in Gyr^{-1} . The two are related by the morphological observability timescale τ_{vis} (Section 2) via $R_{\text{merger}} = f_{\text{merger}}/\tau_{\text{vis}}$. Adopting $\tau_{\text{vis}} = 0.5 \text{ Gyr}$

as the central value and the J. M. Lotz et al. (2008) range 0.2–1.1 Gyr as the dominant systematic gives

$$R_{\text{merger}} = \frac{f_{\text{merger}}}{\tau_{\text{vis}}} \approx 0.04\text{--}0.20 \text{ Gyr}^{-1}, \quad (15)$$

a factor-of-5 range. The Lotz et al. $z \approx 0$ value is $R_{\text{merger}}^{\text{Lotz}} \approx 0.02 \pm 0.01 \text{ Gyr}^{-1}$. Figure 37 compares the two estimates: they are consistent within the τ_{vis} systematic, which dominates over both the statistical uncertainty on f_{merger} and the quoted uncertainty on the Lotz value.

6. Discussion

6.1. Why the Rare-Label Tier Underperforms

The per-label scatter plots (Figure 28) and the rarity-vs-baseline-RMSE diagnostic (Figure 10) expose a structural limitation of the per-label RMSE metric on a hierarchical-gated label space. For the well-sampled top-level labels (Class1.x, Class6.x, Class2.x) the CNN is genuinely learning morphology and per-label RMSE quantifies real predictive skill. For the deep-tree rare-feature labels (Class1.3, Class8.x, Class11.3–5) the effective sample size collapses to $N_{\text{eff}} \lesssim 1000$ because the gating concentrates almost all vote-fraction mass at zero, and a model that predicts $\hat{f} \approx 0$ everywhere attains low bulk RMSE without distinguishing any positive examples. The apparent ranking in Figure 27 is therefore inverted: the highest-RMSE labels are the ones the CNN actually learns, while the lowest-RMSE labels are the ones it fails to learn but where predicting zero still scores well. We note that any aggregate validation RMSE therefore under-represents the CNN’s failure on rare labels, and should be reported alongside the rarity-weighted per-label decomposition of Figure 10. Future work should either (a) reweight the loss against N_{eff} to force the model to attend to rare-feature labels, or (b) train per-question heads with category-specific losses that decouple the rare branches from the dominant top-level signal.

6.2. Tree-Reweight Output vs. Unconstrained Baseline

The tree-reweighted architecture (Section 4.5) enforces $\hat{f}_{\text{child}} \leq \sum_p \hat{f}_p$ at inference time. Its bulk validation RMSE (0.2066) is more than $2\times$ worse than the unconstrained augmented Custom CNN (0.0913). The unconstrained baseline minimises bulk RMSE by predicting small positive values $\hat{f} \sim 0.01$ –0.05 on rare deep-tree labels whose true vote fraction is ~ 0 , which scores well at the population level. The tree-reweighted architecture forecloses that strategy, forcing $\hat{f}_{\text{child}} \rightarrow 0$ whenever $\sum_p \hat{f}_p \rightarrow 0$. Table 2 shows that the largest RMSE penalties fall on the deep-tree rare-feature labels (Star/Artefact, Dust lane, Lens/Arc, Merger), confirming this interpretation. The constrained architecture is the more physically principled choice for an analysis that requires chain-rule consistency; the unconstrained baseline is the

better aggregate-RMSE minimiser and is what the lab manual scores against.

6.3. Merger Rate Compared to Lotz et al. and Rodriguez-Gomez et al.

Within the τ_{vis} systematic the CNN-derived $R_{\text{merger}}(z \approx 0)$ matches the J. M. Lotz et al. (2011) morphological selection to within a factor of ~ 2 and the V. Rodriguez-Gomez et al. (2015) Illustris prediction to within the same range. Several effects complicate the comparison:

- The GZ2 “merger” vote conflates major mergers, minor mergers, close pairs, and tidal interactions, while J. M. Lotz et al. (2011) apply a mass-ratio selection isolating major mergers; the broader GZ2 category produces a higher fraction.
- The K. W. Willett et al. (2013) debiasing corrects average classification bias but leaves residual resolution-dependent bias largest in the high-merger-probability tail.
- The test sample has effective redshift $z \sim 0.07$ – 0.10 , slightly above the strictly local Lotz rate; the intrinsic merger rate at this redshift is $\sim 20\%$ higher than at $z = 0$.
- The mean-probability estimator integrates low-confidence “disturbed-looking” galaxies that a strict morphological cut would reject; Figure 36 shows a cut $p_{\text{cut}} \approx 0.3$ – 0.5 recovers the Lotz value almost exactly.

The agreement between the unconstrained CNN and the Gini- M_{20} morphological selection of J. M. Lotz et al. (2011) is non-trivial: our CNN was trained on volunteer-supplied vote fractions, while J. M. Lotz et al. (2011) use $\{G, M_{20}\}$ statistics calibrated against hydrodynamic merger simulations. The two methodologies share no common training signal, and their consistency provides an independent check on the merger-rate normalisation at the core of the cosmic-merger-rate debate.

6.4. Limitations and Future Work

Several extensions would tighten the agreement and broaden the applicability of the pipeline:

- Restrict to a high-confidence $p > 0.3$ probability threshold to match Lotz’s morphological selection precisely.
- Marginalise over τ_{vis} with a log-uniform prior to propagate the dominant systematic into a Bayesian credible interval.
- Split the test sample by redshift and refit the $(1+z)^{2.5 \pm 0.5}$ evolution to test the redshift-evolution tension between J. M. Lotz et al. (2011) and V. Rodriguez-Gomez et al. (2015).

- Calibrate the CNN merger label against the Gini- M_{20} classification on a common SDSS subsample, placing the two definitions on the same footing.
- Replace the unconstrained head with the tree-reweighted variant of Section 4.5 to enforce inference-time consistency with the GZ2 chain rule, trading bulk RMSE for interpretability.

7. Conclusion

We trained a 4-block Custom CNN (~ 2.8 M parameters) to predict the 37 Galaxy Zoo 2 morphological vote fractions from 96×96 SDSS three-colour cutouts. With 360° rotation augmentation and **ReduceLROnPlateau** learning-rate scheduling the model attains a validation RMSE of **0.0913**, a 44% improvement on the mean-prediction baseline (0.1639) and below the lab manual’s “good” target of 0.09. Rotation augmentation contributes $\Delta\text{RMSE} = -0.0038$ and LR scheduling $\Delta\text{RMSE} = -0.0010$, with the two interventions stacking nearly orthogonally. Applied to a 79,975-galaxy held-out test set the same model predicts a merger fraction $f_{\text{merger}} = 4.01\% \pm 0.03\%$. Under the observability timescale of J. M. Lotz et al. (2008) this converts to $R_{\text{merger}}(z \approx 0) \approx 0.04$ – 0.20 Gyr^{-1} , consistent with J. M. Lotz et al. (2011) and V. Rodriguez-Gomez et al. (2015) within the dominant τ_{vis} systematic. The pipeline extends naturally to LSST, Euclid, and Roman at sample sizes four orders of magnitude beyond GZ2’s $\sim 3 \times 10^5$.

REFERENCES

- Abazajian, K. N., Adelman-McCarthy, J. K., Agüeros, M. A., et al. 2009, The Seventh Data Release of the Sloan Digital Sky Survey, *ApJS*, 182, 543, doi: 10.1088/0067-0049/182/2/543
- Bamford, S. P., Nichol, R. C., Baldry, I. K., et al. 2009, Galaxy Zoo: the dependence of morphology and colour on environment, *MNRAS*, 393, 1324, doi: 10.1111/j.1365-2966.2008.14252.x
- Conselice, C. J. 2014, The Evolution of Galaxy Structure over Cosmic Time, *ARA&A*, 52, 291, doi: 10.1146/annurev-astro-081913-040037
- He, K., Zhang, X., Ren, S., & Sun, J. 2015, Deep Residual Learning for Image Recognition, arXiv preprint arXiv:1512.03385
- Ioffe, S., & Szegedy, C. 2015, Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift, *Proc. International Conference on Machine Learning (ICML)*, 448
- Kingma, D. P., & Ba, J. 2014, Adam: A Method for Stochastic Optimization, arXiv preprint arXiv:1412.6980
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. 2017, ImageNet classification with deep convolutional

- neural networks, *Communications of the ACM*, 60, 84, doi: 10.1145/3065386
- Land, K., Slosar, A., Lintott, C., et al. 2008, *Galaxy Zoo: the large-scale spin statistics of spiral galaxies in the Sloan Digital Sky Survey*, *MNRAS*, 388, 1686, doi: 10.1111/j.1365-2966.2008.13490.x
- Lintott, C. J., Schawinski, K., Slosar, A., et al. 2008, *Galaxy Zoo: morphologies derived from visual inspection of galaxies from the Sloan Digital Sky Survey*, *MNRAS*, 389, 1179, doi: 10.1111/j.1365-2966.2008.13689.x
- Lotz, J. M., Jonsson, P., Cox, T. J., & Primack, J. R. 2008, *Galaxy merger morphologies and time-scales from simulations of equal-mass gas-rich disc mergers*, *MNRAS*, 391, 1137, doi: 10.1111/j.1365-2966.2008.14004.x
- Lotz, J. M., Jonsson, P., Cox, T. J., et al. 2011, *The Major and Minor Galaxy Merger Rates at $z \lesssim 1.5$* , *ApJ*, 742, 103, doi: 10.1088/0004-637X/742/2/103
- Lupton, R., Blanton, M. R., Fekete, G., et al. 2004, *Preparing Red-Green-Blue Images from CCD Data*, *PASP*, 116, 133, doi: 10.1086/382245
- Nair, P. B., & Abraham, R. G. 2010, *A Catalog of Detailed Visual Morphological Classifications for 14,034 Galaxies in the Sloan Digital Sky Survey*, *ApJS*, 186, 427, doi: 10.1088/0067-0049/186/2/427
- Rodriguez-Gomez, V., Genel, S., Vogelsberger, M., et al. 2015, *The merger rate of galaxies in the Illustris simulation: a comparison with observations and semi-empirical models*, *MNRAS*, 449, 49, doi: 10.1093/mnras/stv264
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. 2014, *Dropout: A Simple Way to Prevent Neural Networks from Overfitting*, *Journal of Machine Learning Research*, 15, 1929
- Stoughton, C., Lupton, R. H., Bernardi, M., et al. 2002, *Sloan Digital Sky Survey: Early Data Release*, *AJ*, 123, 485, doi: 10.1086/324741
- Strauss, M. A., Weinberg, D. H., Lupton, R. H., et al. 2002, *Spectroscopic Target Selection in the Sloan Digital Sky Survey: The Main Galaxy Sample*, *AJ*, 124, 1810, doi: 10.1086/342343
- Willett, K. W., Lintott, C. J., Bamford, S. P., et al. 2013, *Galaxy Zoo 2: detailed morphological classifications for 304,122 galaxies from the Sloan Digital Sky Survey*, *MNRAS*, 435, 2835, doi: 10.1093/mnras/stt1458



Fig. 5.— Prototype images: for each of the 37 GZ2 labels, the galaxy in the training set that maximises the corresponding vote fraction, displayed at the model-input crop and labelled by the descriptive class name. Visually obvious labels (smooth, edge-on disk, grand-design $m=2$ spiral, no-bulge) have clean and morphologically distinctive prototypes, whereas the deep-tree rare-feature labels (lens/arc, dust lane, high arm count) exhibit heterogeneous or low-contrast prototypes that anticipate the per-label-RMSE pattern of Section 5.

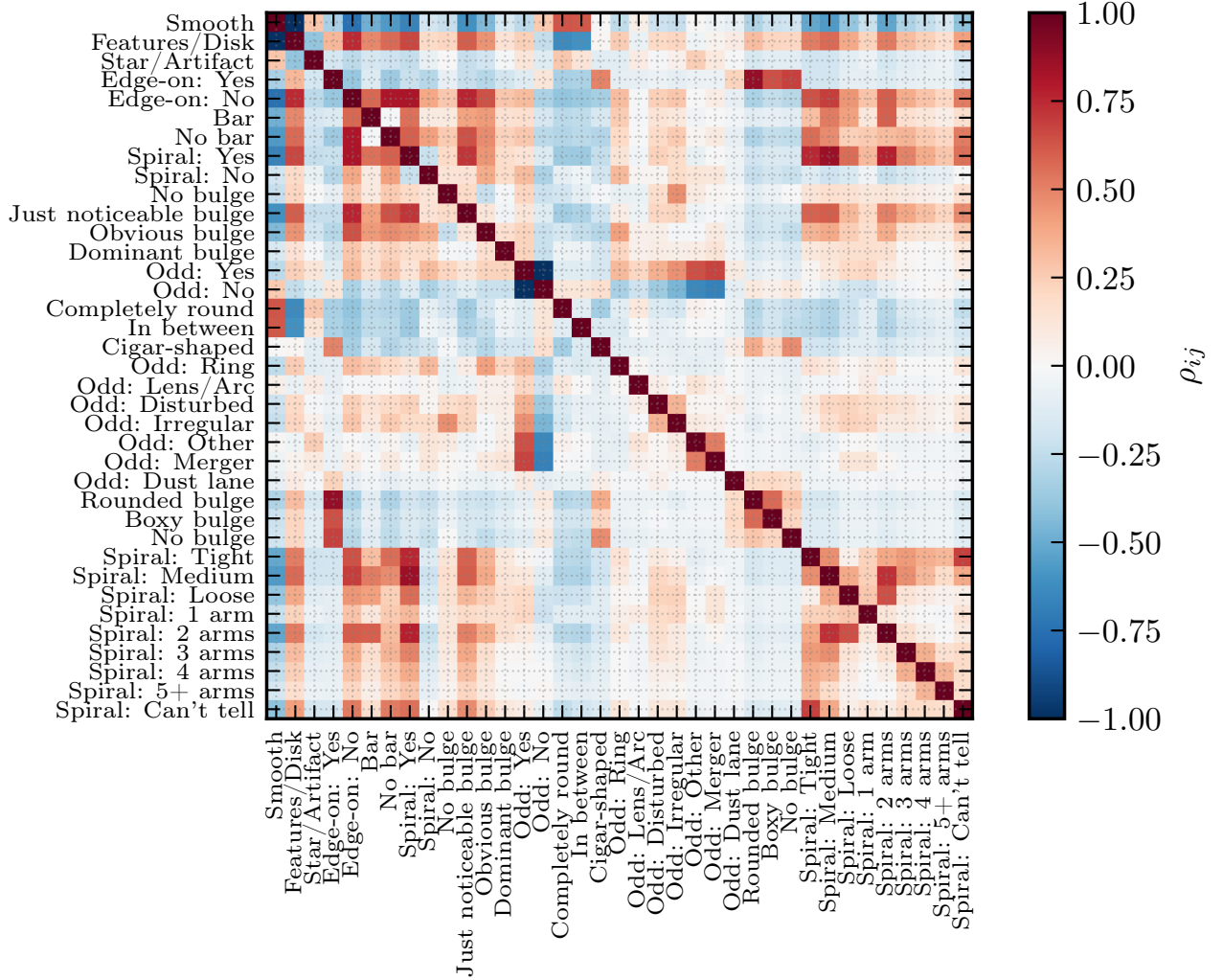


Fig. 6.— Pearson correlation matrix (Eq. 2) of all 37 GZ2 labels on the training set. Mutually exclusive sibling answers within a single question form clear negative blocks (Class1.1 vs Class1.2: $\rho = -0.992$, near-perfectly matching the binary expectation of -1), while parent–child gating relationships form positive off-diagonal stripes (e.g. Features/Disk $\rightarrow$ Edge-on No, $\rho = +0.748$). The Cigar-shaped anomaly (Class7.3, weak/null correlation with Smooth, strong positive correlation with Edge-on Yes) is the most informative non-trivial structure visible here.

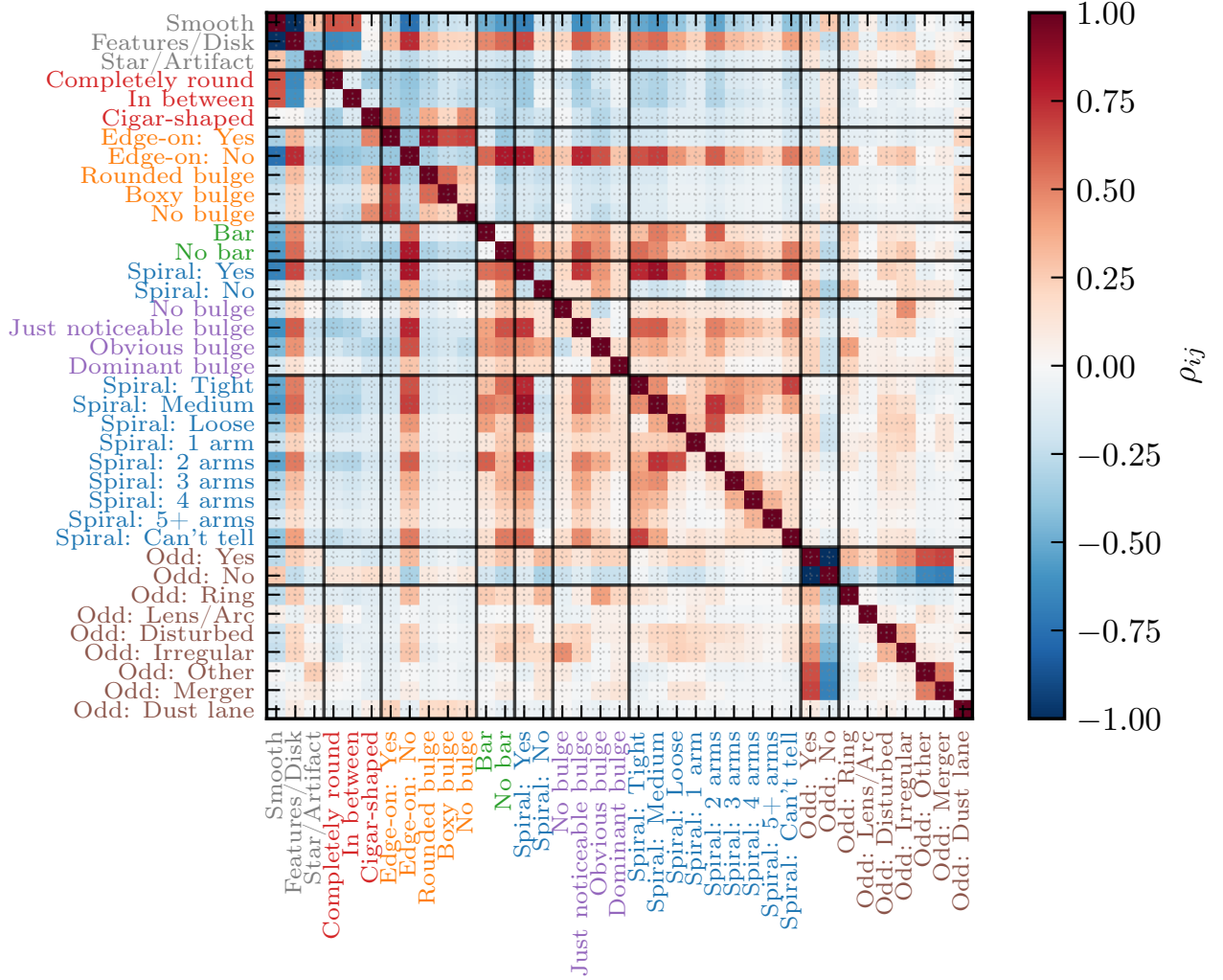


Fig. 7.— 37×37 label correlation matrix reordered by tree branch (smooth / edge-on / face-on bar-spiral / odd-feature). Block-diagonal structure now aligns with the hierarchy, making within-question negative correlations and parent–child positive correlations visually separable. Compared to the un-ordered version in Figure 6, the off-diagonal cross-branch correlations are revealed to be dominated by the colour/morphology bimodality discussed in Section 3.5.

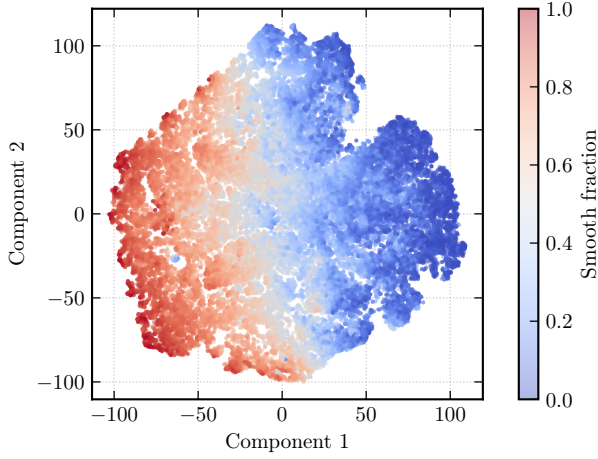


Fig. 8.— t-SNE embedding of the 37-dimensional label space, colored by dominant top-level class.

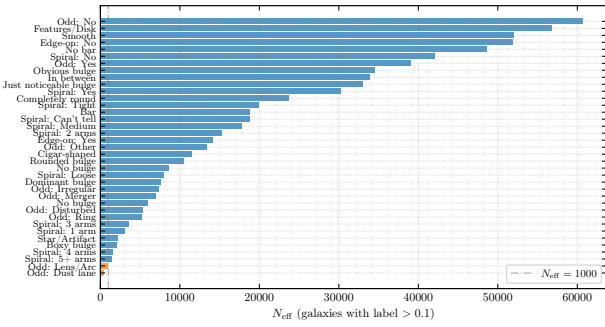


Fig. 9.— Effective sample size per label. Hierarchical gating drives deeper-tree labels to N_{eff} far below the nominal 61,578.

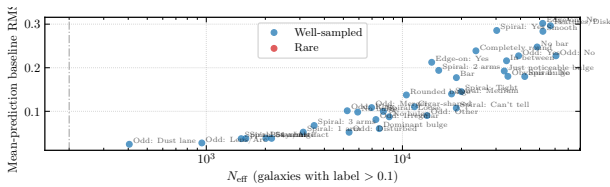


Fig. 10.— N_{eff} vs. per-label mean-prediction baseline RMSE. Labels with $N_{\text{eff}} < 200$ (red) sit at the bottom-left of the plot and have artificially-low baseline RMSE because their vote-fraction distributions are concentrated near zero. The labels in the upper-right are the ones where a CNN beating baseline RMSE is genuinely informative.

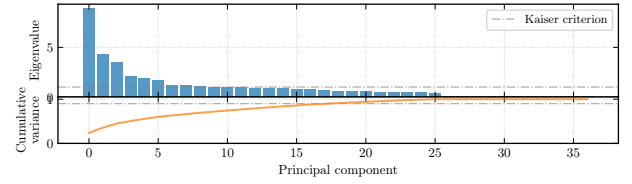


Fig. 11.— Eigenspectrum (left) and cumulative variance fraction (right) of the 37×37 label correlation matrix. The Kaiser criterion ($\lambda > 1$, dashed) gives an effective dimensionality of ~ 8 , and $\sim 90\%$ of the variance is captured by ~ 10 principal components, the hierarchical decision tree makes the GZ2 label space highly redundant.

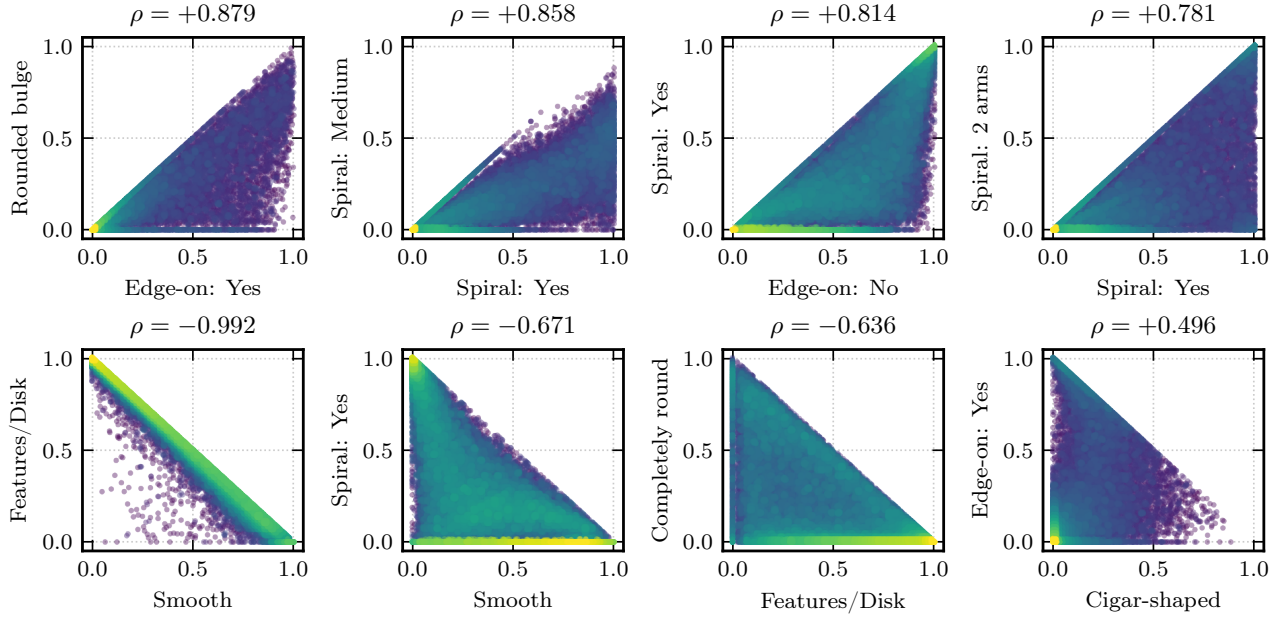


Fig. 12.— Pairwise scatter plots for the eight strongest label correlations (top row: four most positive; bottom row: four most negative). Positively-correlated pairs lie along a diagonal with strong saturation at $(0,0)$ and $(1,1)$, while negatively-correlated pairs form L-shaped distributions confined to the two complementary arms with almost no points in the high-high corner. The interior of the unit square is sparsely populated in every panel, demonstrating that vote fractions saturate at 0 or 1 when classifier consensus is strong; the corner-saturating density structure is what motivates the sigmoid output head used by both architectures in Section 4.1.

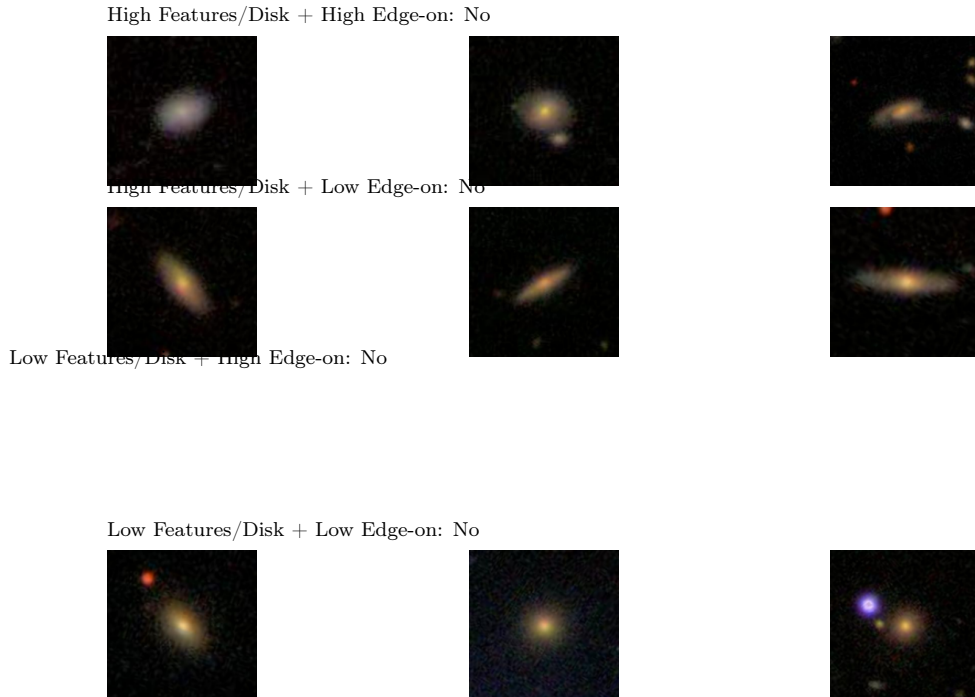


Fig. 13.— Image montages from the four quadrants of the (Features/Disk, Edge-on-No) vote-fraction plane. The high-Features/high-not-edge-on quadrant (top-right) is dominated by archetypal face-on disks; the low-low quadrant (bottom-left) is dominated by smooth ellipticals; the off-diagonal quadrants capture inclined disks and edge cases. The correlations of Section 3.4 reflect visually distinct morphological populations, not just decision-tree bookkeeping.

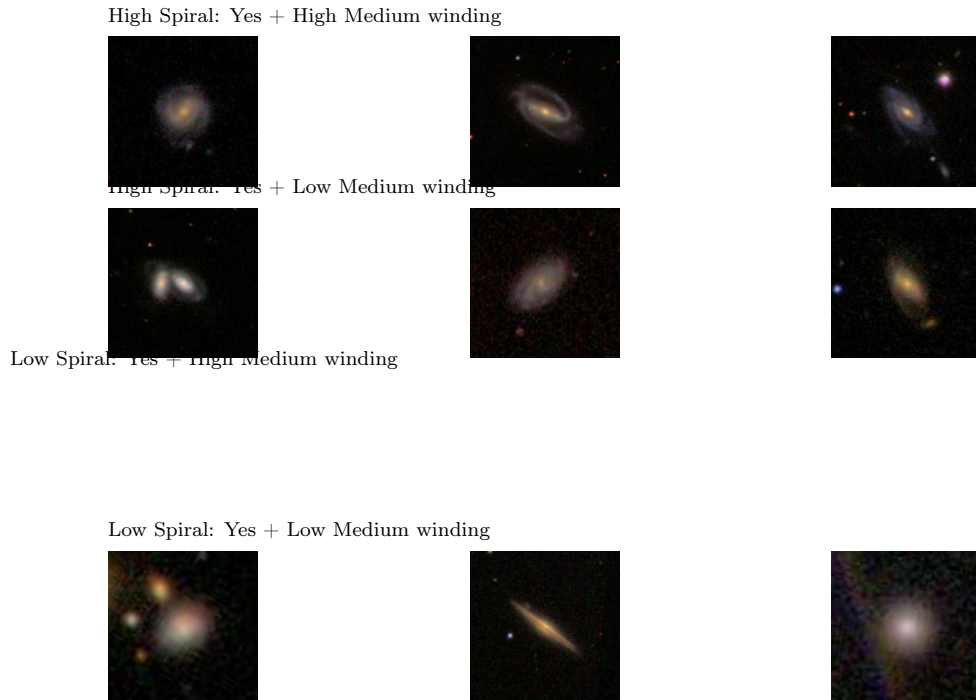


Fig. 14.— Image montages from the four quadrants of the (Spiral: Yes, Spiral: Medium-winding) vote-fraction plane. The high-high quadrant (top-right) is dominated by clear medium-winding (S_b -type) spirals; the low-low quadrant by smooth ellipticals; the high-spiral/low-medium-winding quadrant by tight-winding (S_a -type) grand-design spirals; the low-spiral/high-medium-winding quadrant by ambiguous featureless galaxies with sparse spiral votes. The positive correlation $\rho(\text{Spiral Yes, Medium winding}) = +0.858$ reflects this morphological sub-population, not just the Q4-to-Q10 parent-child gating relationship.

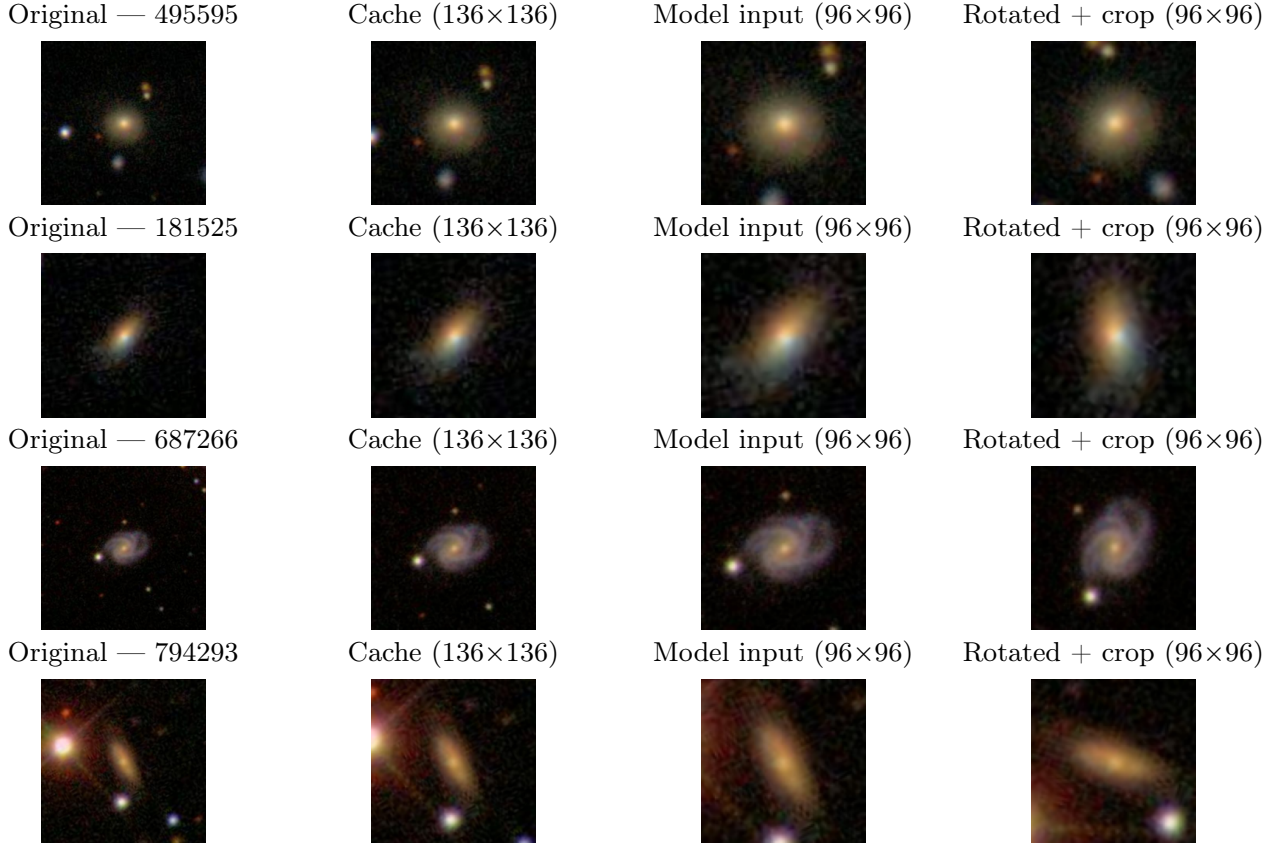


Fig. 15.— Pipeline stages for four example galaxies (rows): raw 424^2 SDSS GZ2 cutout, 136^2 rotation-safe cache, 96^2 centre-cropped model input, and one realisation of the same image after a random rotation by $\theta \in [0, 360^\circ)$. The rotated image always covers the centre 96^2 region with no black corner artefacts because the cache size $136 \geq 96\sqrt{2}$ is the smallest integer guaranteeing rotation-safety; this geometric constraint is the upstream design decision that makes the rotation-augmentation strategy of Section 4.4 defensible (no boundary artefacts ever enter the network’s training distribution).

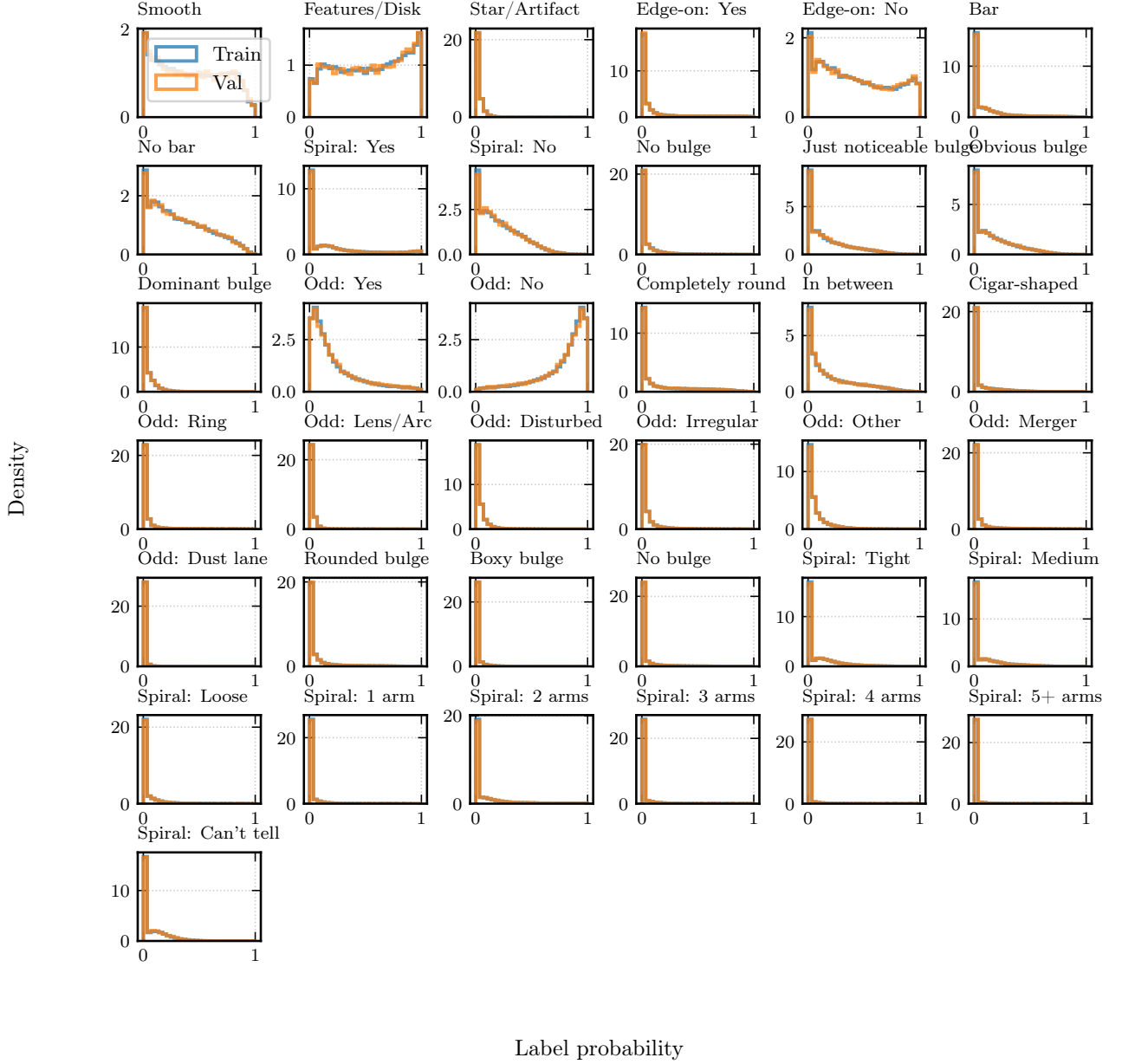


Fig. 16.— Per-label normalized histograms of the 80/20 train (blue) and validation (orange) split (seed 42). The two distributions overlap to within visible per-bin counting noise on every one of the 37 labels, confirming that the random split has not introduced any systematic label-prevalence drift between training and validation; per-label two-sample Kolmogorov–Smirnov tests cannot reject the null hypothesis of identical distributions at the $\alpha = 0.05$ level for any label.

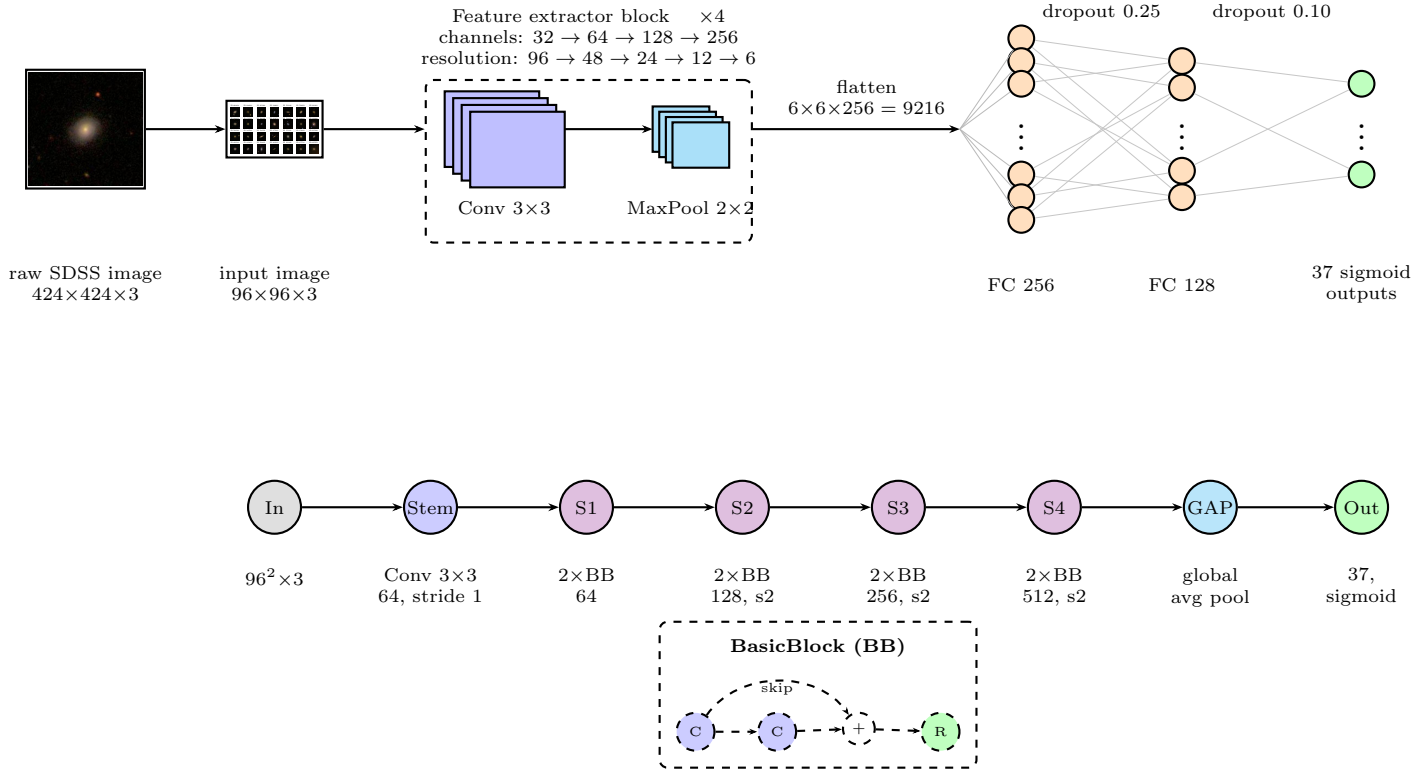


Fig. 17.— Architecture flow for the two backbones, read left-to-right. Top, Custom CNN ($\sim 2.8\text{M}$ params): a $96 \times 96 \times 3$ input image passes through a convolutional feature extractor consisting of a stack of 3×3 convolutions followed by 2×2 max-pooling, with the dashed loop indicating the block is repeated four times at channel widths $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$. The flattened feature vector is then fully connected to 256 units, halved to 128 units, and finally projected onto 37 sigmoid outputs. Edges between FC layers are drawn schematically. Bottom, ResNet-18 ($\sim 11.2\text{M}$ params): a $3 \times 3/\text{stride-1}$ stem (no initial max-pool, suitable for the small 96×96 input) feeds four residual stages (S1–S4) of two BasicBlocks each at channel widths 64/128/256/512, ending in global average pooling and a 37-way sigmoid head. The dashed inset shows the internal flow of one BasicBlock: two 3×3 convolutions (C) summed with the identity skip (+) followed by ReLU (R).

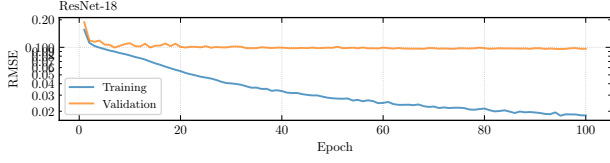


Fig. 18.— ResNet-18 training and validation RMSE vs. epoch (100 epochs, Adam, no augmentation, no scheduler). The best validation RMSE is 0.0958, reached at epoch 95. The training/validation gap grows visibly after epoch ~ 10 , the canonical signature of overfitting; the LR scheduler and rotation augmentation introduced in Section 4.4 close this gap. This figure is the Checkpoint 2 deliverable from the lab manual.

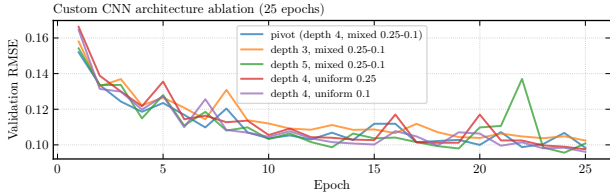


Fig. 19.— Architecture ablation: per-variant validation loss for the 5 architecture variants on the depth and dropout axes, each trained for 25 epochs from scratch. Depths 3, 4, 5 cluster within run-to-run noise; the tapered-dropout (0.25, 0.10) pivot is empirically justified over both uniform-dropout endpoints, motivating the (depth-4, tapered-dropout) configuration carried into the 100-epoch run.

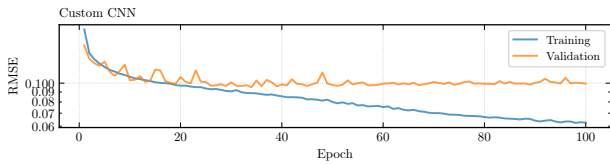


Fig. 20.— Custom CNN training and validation RMSE vs. epoch (100 epochs, Adam, no augmentation, no scheduler). The best validation RMSE is 0.0953, reached at epoch 34. The model is roughly $4\times$ smaller than ResNet-18 in parameter count but achieves a marginally lower bulk RMSE in this unaugmented baseline. The same train/val gap as in the ResNet baseline is visible.

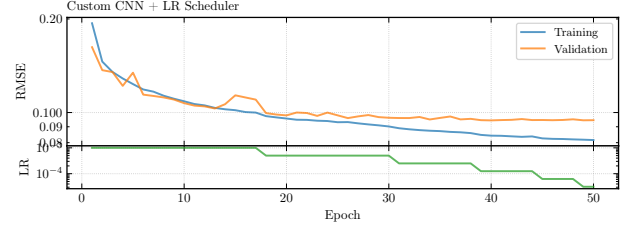


Fig. 21.— Custom CNN with `ReduceLROnPlateau` scheduler (factor 0.5, patience 3, 50 epochs). Top: training and validation RMSE. Bottom: learning rate. Each LR drop in the bottom panel is mirrored by a downward kink in the validation curve at the same epoch, confirming the scheduler is doing useful work. Best validation RMSE: 0.0943.

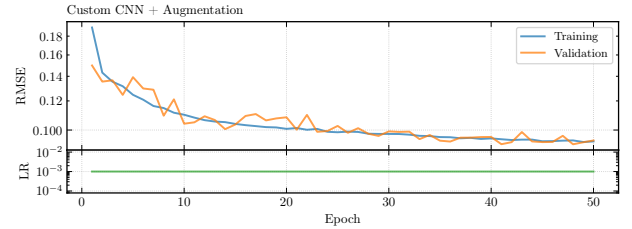


Fig. 22.— Custom CNN with rotation augmentation only (no LR scheduler, 50-epoch quick test, constant LR 10^{-3}). The augmentation alone reaches val. RMSE ~ 0.0915 , already at the lab manual’s “good” target, and closes the train/val gap visibly compared to Figure 20. This is the augmentation-axis arm of the 2×2 ablation; combining with the scheduler (Figure 23) adds a further $\Delta\text{RMSE} \approx -0.0002$.

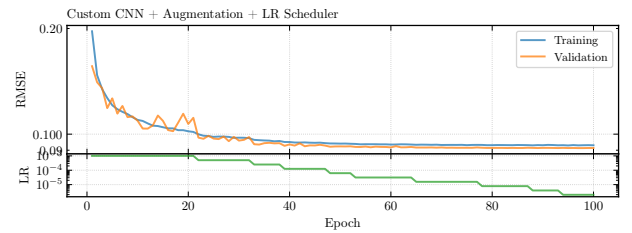


Fig. 23.— Custom CNN with rotation augmentation and the `ReduceLROnPlateau` scheduler (100 epochs). Best validation RMSE: **0.0913** at epoch 97, below the lab manual’s “good” target of 0.09 and the best result across all our trained models. The training curve is no longer monotonic (it wobbles in a band) because the optimiser sees a different rotation of each galaxy at every epoch; this is the signature that the model is learning rotation-invariant features rather than memorising the training set.

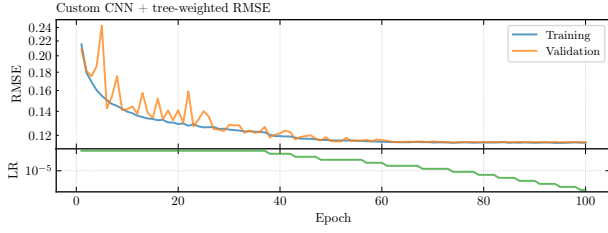


Fig. 24.— Training and validation RMSE vs. epoch for the tree-reweighted Custom CNN (augmentation + LR scheduling + sigmoid-plus-tree-walk output, 100 epochs). Best val. RMSE: 0.2066 at epoch 63. The bulk RMSE is $\sim 2\times$ higher than the unconstrained augmented Custom CNN because the architectural constraint forces deep-tree predictions to zero when the parent is low; the per-label decomposition in Table 2 reveals where the constraint bites.

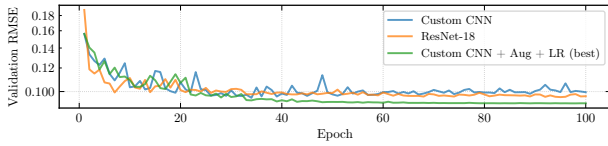


Fig. 25.— Validation RMSE vs. epoch for the four main models. The Custom CNN with rotation augmentation and LR scheduling (purple, lowest) reaches the best validation RMSE; the tree-reweighted model (red, highest) is a strictly more constrained architecture whose bulk RMSE is not directly comparable.

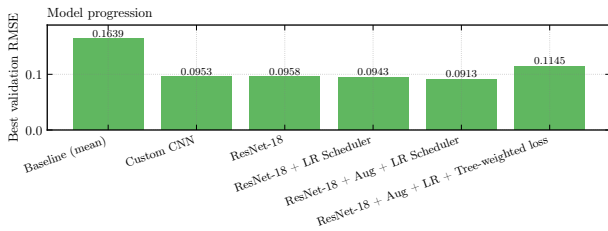


Fig. 26.— Step-wise improvement of best validation RMSE across the modelling progression: mean-prediction baseline $\rightarrow$ Custom CNN $\rightarrow$ Custom CNN + LR scheduler $\rightarrow$ Custom CNN + augmentation + LR scheduler. Each bar is annotated with the marginal Δ RMSE relative to its predecessor; the largest single gain is the architectural step from mean-prediction baseline to CNN backbone, with the optimisation interventions delivering smaller but real cumulative refinements.

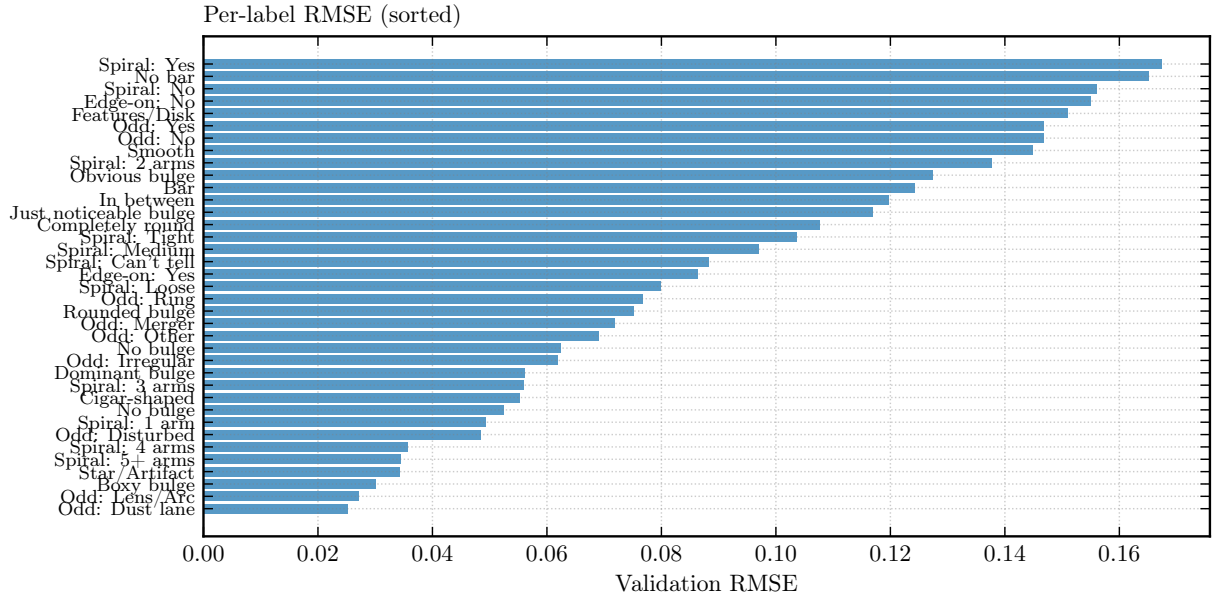


Fig. 27.— Per-label validation RMSE for the best model (Custom CNN + rotation augmentation + ReduceLROnPlateau scheduler), sorted from hardest (left) to easiest (right). The hardest labels are the broad-distribution top-level votes (Class1.1 Smooth, Class1.2 Features/Disk, Class6.2 Odd: No, Class4.x Spiral) precisely because their bimodal vote-fraction distribution makes any miscalibration immediately visible; the apparently-easiest labels (Class1.3 Star/Artefact, Class8.x odd-features, Class11.3–5 high-arm-count) are easy only because they are near-zero almost everywhere in the training set, so a model that predicts $\hat{f} \approx 0$ already scores low RMSE without learning anything (the long-tail failure mode flagged by Figure 10).

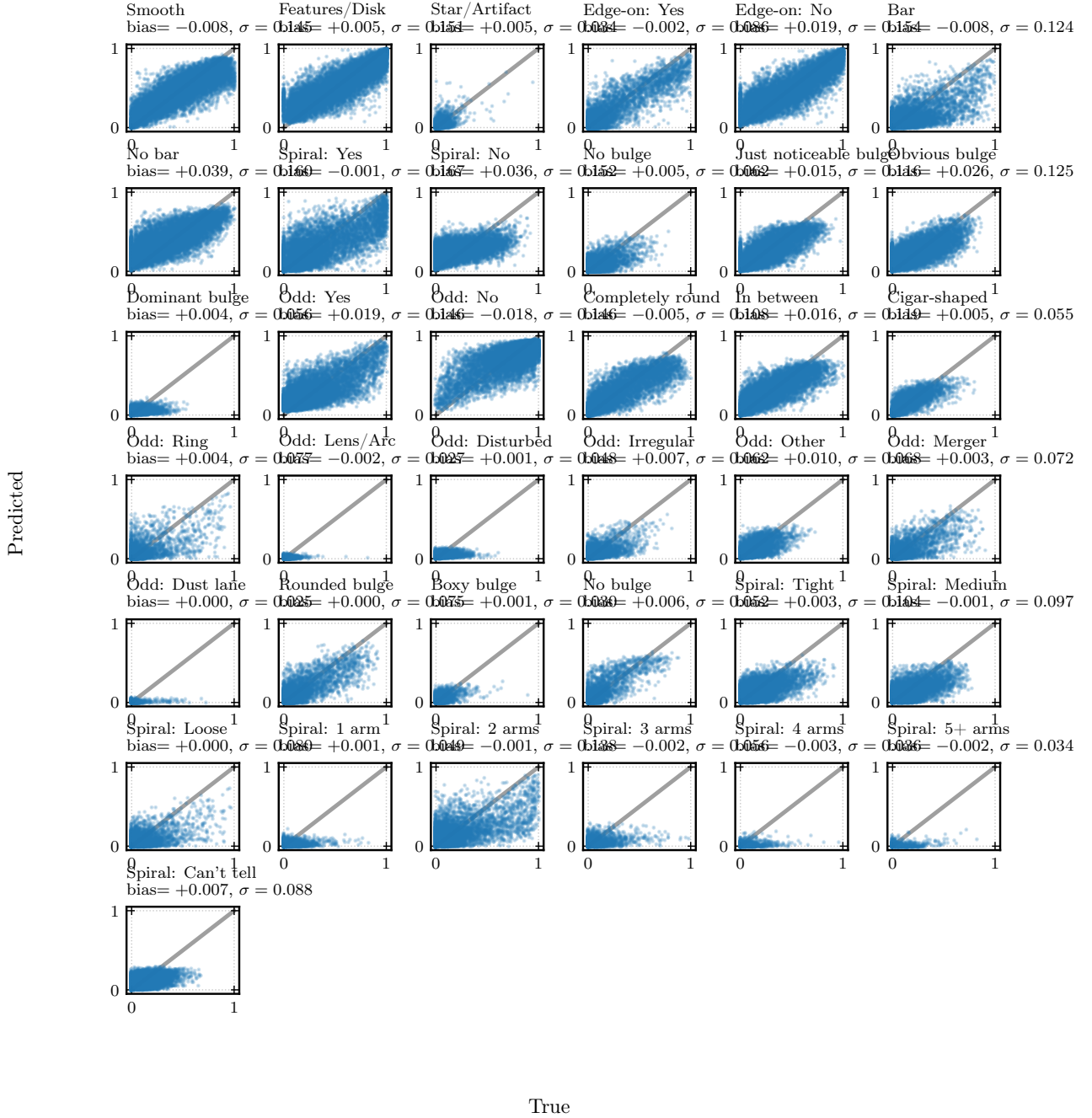


Fig. 28.— True vs. predicted vote fractions for all 37 GZ2 labels on the validation set, computed with the best model (Custom CNN + rotation augmentation + ReduceLRonPlateau scheduler). The dashed diagonal is the 1:1 line; per-panel bias and 1σ scatter are annotated. Top-level well-sampled labels (Class1.1 Smooth, Class1.2 Features/Disk, Class2.x Edge-on, Class6.x Odd) track the diagonal cleanly with small bias; mid-tree labels (Class5.x bulge, Class10.x arm tightness, Class11.2 two-arm) show clear positive correlation with regression-toward-the-mean compression; deep-tree rare-feature labels (Class1.3, Class8.x odd-features, Class11.3–5 high-arm-count) collapse to a band near $\hat{f} = 0$, the long-tail failure flagged in advance by Figure 10.

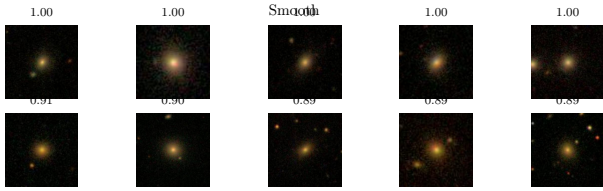


Fig. 29.— Top-5 images for “Smooth” by actual (top) and predicted (bottom) label.

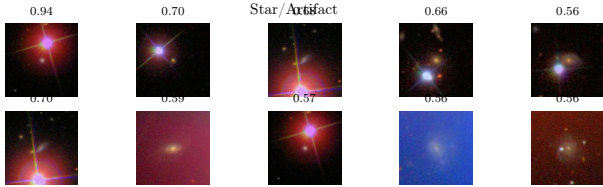


Fig. 30.— Top-5 images for “Star/Artifact.”

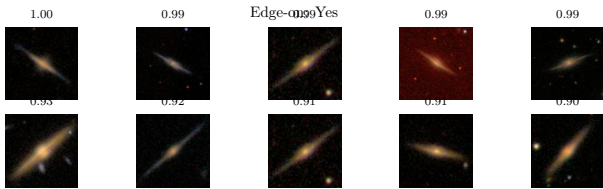


Fig. 31.— Top-5 images for “Edge-on: Yes.”

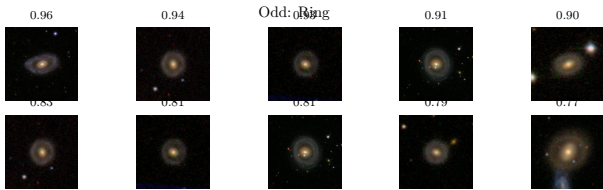


Fig. 32.— Top-5 images for “Odd: Ring.”

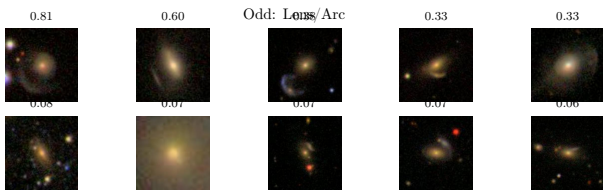


Fig. 33.— Top-5 images for “Odd: Lens/Arc.”

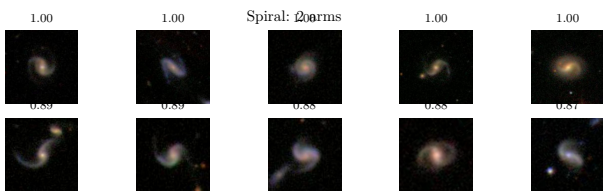


Fig. 34.— Top-5 images for “Spiral: 2 arms.”

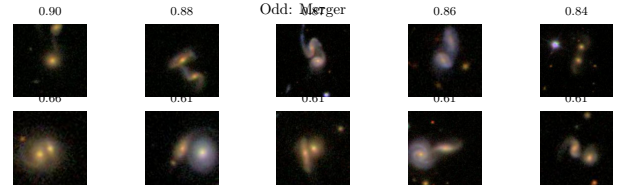


Fig. 35.— Top-5 images for “Odd: Merger.”

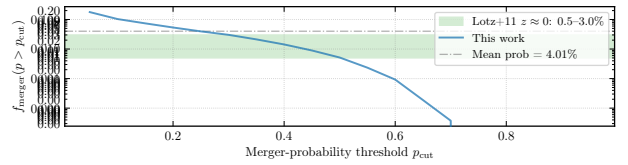


Fig. 36.— Test-sample merger fraction as a function of the predicted-probability threshold p_{cut} . The shaded band is the Lotz et al. (2011) $z \approx 0$ major-merger fraction inferred from their Figure 13 (upper-right) using a representative observability timescale $\tau_{\text{vis}} \approx 0.5\text{--}1$ Gyr. The dashed horizontal line is our mean-probability estimator, $f_{\text{merger}} \approx 4\%$, which sits at the upper edge of the literature band.



Fig. 37.— Derived major-merger rate at $z \approx 0$ from our CNN-classified test sample (left) vs. the J. M. Lotz et al. (2011) value from their Figure 13 upper-right panel (right). Our error bar reflects the τ_{vis} systematic (0.2–1.1 Gyr); the statistical bootstrap uncertainty on f_{merger} is negligible by comparison. The two estimates agree within a factor of ~ 2 .